Degree-separated arithmetic encodings and an ordinary-root merger for Con(ZFC)

OpenAI-assisted research for Vladimir Reshetnikov and ProveIt

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Abstract

We review all eight newly delivered wave-seven packages and combine their ideas with a different coefficient-extraction architecture. Quadratic, linear and constant source terms are encoded in separate kernels, so their labels may collide across degrees without corrupting a tested coefficient. This reduces the source convolution height from 5556 to 2161. Combined with calibrated carry blocks, exact CRT parametrization and sharper Pell majorants, it yields smaller explicit nine-natural and ten-integer polynomials. A coprime spectator guard removes an unnecessary radical weight in the thirteen-integer route. Independently, retaining the ordinary Pell root rather than its multiplier yields a twelve-positive-variable polynomial of degree 316. A late composition with a parallel free-coordinate construction gives the recommended nine-positive equation at degree 1 910 404 and 473 operations. All coefficients are fixed by explicit recipes and materialized native payloads. The proofs are ordinary mathematical reductions with identified inherited dependencies, accompanied by finite exact certificates; no consistency proof, global minimum or new end-to-end formal-kernel certificate is asserted.

1 Question and reproducible review boundary

The task is to construct an explicit integer-coefficient polynomial F for which ZFC proves

$$(\neg\exists \mathbf{x} \in D^r \; F(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con(ZFC)},$$

with a small description, low degree, few unknowns or a short arithmetic circuit. These are different size objectives. We state the domain D explicitly: $\mathbb{N}$ includes zero, positive integers exclude zero, and unrestricted integers form a third domain. A positive-domain equation is not silently reinterpreted over $\mathbb{Z}$.

The eight input archives were retrieved at arrival commit `4b4349064a75`, verified against their Git blob hashes, and inspected as original delivered packages. Their placement commit `6d375f2d2dea` assigns sources 46–53 under

https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

Full commit IDs, archive hashes and member provenance are in `INCOMING_PINS.json` and the component ledgers. The catalogue was not yet reconciled at this placement, so comparisons use the original packets rather than an outdated summary. Source53 is the preceding 32-page delivery in this conversation.

Every register below abbreviates a polynomial expression, not an additional existential unknown. All exponents in the final circuits are fixed nonnegative integers. Rational quantities, square roots, valuations and divisibility appear in the proofs, not as permitted operations in the polynomial. The route is finite ZFC refutation $\leftrightarrow$ calibrated blank K215 halting $\leftrightarrow$ a native-window/Pell source solution $\leftrightarrow$ a zero of the explicit new polynomial. Negation supplies the requested equivalence without deciding either side.

2 Results against the pinned eight-packet snapshot

A charged operation is a binary addition, subtraction or multiplication, including scalar multiplication and all multiplications implementing a fixed power. Shared registers and fixed coefficient leaves are not charged again. The bit complexity of the huge fixed integers is not hidden inside this circuit metric; their recipes and payloads are supplied separately.

Domain and unknowns	Exact degree	Operations	Relevant prior point
9 positive	1 910 404	473	Source53: 7 344 328/738
$10\mathbb{Z}$	8 469 528	533	Source53: 21 111 400/659
$13\mathbb{Z}$, compact Jones	11 543 632	518	Source53: 57 866 184/759
12 positive	316	102	Source52: 897/108
14 positive	52	79	Source50: 56/79
15 positive	125	67	Original 155/67

The recommended nine-positive row includes the late free-coordinate composition proved in the final supplement; the earlier degree 2 676 548/476 evaluator is retained for comparison. The thirteen-integer row uses the proved compact-calibration companion; the main positive-threshold presentation has degree 11 546 112 and 520 operations. The nine-natural version of the primary equation uses 480 operations at the same degree. The twelve-positive system has residual degrees $(2, 26, 288)$ with 95 shared operations, or 94 after the justified positive transport shift. The sum-of-squares alternative has degree 576 and 103 operations. These systems are secondary presentations; the main rows above are single polynomial equations.

The new nine- and ten-variable primary points improve not just the preceding package in this conversation but both extremes of the reviewed frontier. Source52's lowest-cost nine-positive equation uses 548 operations at degree 15 502 748; its lowest-cost ten-integer equation uses 582 operations at degree 50 822 936. The new equations have both smaller degree and fewer operations than those points. A retained nine-positive comparison uses 475 operations at degree 9 336 884, trading one gate for a substantially larger degree and using its separately bound complete native tuple.

The old unknown counts are not claimed as new universal representation bounds. Sun's 2026 result is used for the ten-unrestricted-integer equation only. The nine-natural, thirteen-integer, twelve-positive and short forms use their stated Jones/Pell routes without that theorem. The twelve-positive polynomial is especially concise at moderate degree, but it does not beat the source52 eleven-positive unknown count: the latter remains degree 954 at 113 operations.

Short-circuit tradeoffs. Combining the squared-modulus quotient with the ordinary strong norm and root-gap coordinates gives the following fifteen-positive points, listed as (operations, exact degree):

$$(67, 125),\ (68, 101),\ (69, 91),\ (70, 84),\ (71, 80),$$
$$(72, 74),\ (73, 60),\ (74, 50),\ (75, 46).$$

The historical $(76, 44)$ point is included only as a reference. The source51 absolute operation record $(66, 157)$ remains unbeaten; we do not claim a sub-66 circuit. Thus there is no single equation that minimizes unknowns, degree, operations and coefficient storage simultaneously.

What was reviewed. Sources46–49 contribute shared positive-coordinate words, selective support codes and sharper radical majorants. Source50 supplies midpoint/index absorption and the ordinary-strong fourteen-variable bootstrap. Source51 supplies squared-modulus quotient reconstruction and the 66-operation record. Source52 combines CRT removal, positivity-first separation, better majorants and lean direct Pell carriers. Source53 supplies our previous calibrated carry and seeded-guard constructions.

We inspected every entry point and the source proofs used here; we did not claim to rerun every upstream archive wholesale. New local data and circuit properties are recomputed by this release's independent finite audits.

Coherent coefficients. The initial six families use the complete source31 Wang943 121×8801 tuple with $B_{\text{native}} = 2^{1064921}$, $\beta = B_{\text{native}} - 1$, and its matching K, M_C, M_F. The last is the already shifted high mask. The four uncompressed payloads total 533,099 bytes. The Sun and signed source floor is $2^{2134940} = 8192 \cdot 2^{2134927}$; the nine-variable source uses $2^{2134939} = 4096 \cdot 2^{2134927}$, with its own strict coefficient and calibration bounds. No port from the smaller 2^{590069} tuple is spliced into this one. That smaller whole tuple appears only in explicitly marked comparison circuits. We claim no new coefficient byte record and no expanded-polynomial storage advantage from treating fixed leaves as circuit inputs.

3 What changes mathematically

The previous selective code still homogenized a quadratic system by adding a dummy coordinate. It therefore separated linear from quadratic labels although no source equation needed those cross-degree products. We instead use

$$G(B, U) = D_2(B)U^2 + D_1(B)U + D_0(B).$$

Only labels within the same homogeneous degree can interfere. The resulting finite certificate has 89 source records, 22 exact target identities and 22,544 interactions, rather than the preceding 136 records and 71,808 interactions. It deliberately permits a linear label to equal a quadratic label. A signed leading-layer estimate is needed: positive input coordinates do not by themselves make a signed kernel positive.

The arithmetic finalizers also impose only the relations actually needed. The mask's binary population calibrates the binomial divisibility threshold; a small factor in the block length makes that threshold a polynomial square. An exact CRT parametrization removes one divisibility quotient. A spectator radicand in the last Pell guard prevents coincident radicals without a large external separator. Finally, choosing which Pell witness to retain changes the direct moderate-degree construction dramatically: keeping f makes $Q = \Delta(f^2 - 1)$ degree 14 instead of the normalized degree 46 expression. The new merger restores integrality before invoking strong-rank results.

4 Degree-separated source extraction

The gain in this section comes from changing the multiplication architecture, not from finding a smaller code for the previous homogenized square. Linear and quadratic monomials are placed in separate kernels, so that their labels may coincide. For the explicit code below the linear monomial N and the quadratic monomial J^2 both have label 2; this would be prohibited by every previous dummy-coordinate isolation condition.

4.1 Source and exact finite identity

Let $r_1, \ldots, r_{22}$ be the unshifted residuals in the original 31-positive-variable package-31 source, with the complete original four-port tuple (β, K_0, M_C, M_F); M_F already includes its prescribed shift. First put

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_j = r_j \quad (j \neq 14, 15).$$

The inverse operations $r_{14} = \rho_{14} - \rho_2$ and $r_{15} = \rho_{15} - \rho_3$ prove equality of the common zero sets on every domain. There are now 89 records: 7 constants, 39 linear records, and 43 quadratic records. The support union comprises 1, 22, and 39 monomials in the respective degrees.

Write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For positive, distinct integer labels w_i, set

$$U(Z) = \sum_{i=1}^{31} p_i Z^{w_i}, \qquad e(\alpha) = \sum_i \alpha_i w_i,$$

$$c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & |\alpha| = 0 \text{ or } 1, \\ a_{j,\alpha} \prod_i \alpha_i!, & |\alpha| = 2, \end{cases} \qquad D_d(Z) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} Z^{q_j - e(\alpha)}.$$

Our packed expression is

$$G(Z,p) = D_2(Z)U(Z)^2 + D_1(Z)U(Z) + D_0(Z). \tag{1}$$

All exponents in the three kernels are nonnegative.

The finite certificate imposes two conditions, separately for $d = 0, 1, 2$. Every supported degree-d monomial must have a unique label among all monomials of that degree. Furthermore, a source term of D_d from row j can reach a marked target q_k after multiplication by a degree-d monomial only when $k = j$ and the monomials agree. There are respectively 1, 31, and 496 candidate monomials. The coefficient of a cross product in U^2 is 2 and that of a square is 1, so the factorial weighting gives

$$[Z^{q_j}]G(Z,p) = 2\rho_j(p) \qquad (1 \le j \le 22). \tag{2}$$

This is a universal identity in the 31 witness indeterminates and the four ports. Cross-degree collisions cause no interference: different homogeneous degrees never multiply the same kernel.

The labels, in the source variable order, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\text{aux}}, \Delta),$$
$$(1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,$$
$$2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

Assign targets in the following row order:

$$(2, 22, 8, 15, 9, 11, 3, 17, 13, 18, 12, 20, 4, 21, 19, 1, 6, 10, 16, 14, 7, 5),$$
$$(28, 85, 286, 384, 593, 602, 642, 731, 906, 966, 1083, 1212, 1231,$$
$$1393, 1461, 1471, 1678, 1719, 1732, 1880, 2107, 2129).$$

The exact sparse kernels are supplied as machine-readable polynomials in `split_kernels.json`, derived from the canonical `split_code.json`. The checker independently parses the original residuals, proves the two inverse row identities, enumerates 22 544 source/expansion products, and compares all 22 formal output polynomials. There are exactly 89 target contributions. It obtains

$$M = 116, \quad Q = 2129, \qquad (\deg D_0, \deg D_1, \deg D_2) = (1719, 2027, 1929),$$

$$H = \max_{0 \le d \le 2} (\deg D_d + dM) = 2161. \tag{3}$$

Every kernel has internally distinct exponents. The leading term of D_2 is exactly $+Z^{1929}$, arising from the EX term in row 5.

4.2 Coefficient bounds and unconditional positivity

The complete native tuple has strict port bounds

$$\beta < 2^{1064921}, \quad K_0 < 2^{1070004}, \quad M_C < 2^{1064921}, \quad M_F < 2^{1064922}.$$

Taking absolute integer coefficients and replacing the port powers by their bit bounds yields, for every weighted kernel coefficient, absolute value strictly below

$$C = 2^{2134927}.$$

The checker recomputes this finite integer majorant from every coefficient. The common selected source floor $\beta_S = 2^{2134940}$ satisfies $\beta_S > 4096C$, so $B = \beta_S b^2 > 4096Cb^2$. For the Jones choice $B = b^3$, it suffices to retain the existing requirement $b \geq 4096C$.

Put $\ell = \sum_i B^{w_i}$ and $U = \ell + g$. If $g \geq 0$, then $U \geq B^{116}$. For $B \geq 4096C$, the certified leading coefficient gives

$$D_2(B) > B^{1929}\left(1 - \frac{C}{B-1}\right) > \frac{1}{2}B^{1929},$$

whereas $|D_1(B)| < 2CB^{2027}$ and $|D_0(B)| < 2CB^{1719}$. After division by $B^{1929}U^2$, the two potentially negative terms are less than $2C/B^{18} < 1/8$ and $2C/B^{442} < 1/8$. Thus

$$G(B,U) > \frac{1}{4}B^{1929}U^2 > 0 \qquad (g \geq 0). \tag{4}$$

This proof precedes digit recovery and does not assume that B is already a power of two. In particular, it does not incorrectly infer positivity from the signs of all kernel coefficients.

4.3 Sparse padding and the source interval

Suppose $b \geq 4$, $B \geq 4096Cb^2$, and $0 \leq g < 2bB^{116}$. Distinct positive labels imply $\ell < 2B^{116}$, whence $U < 3bB^{116}$. By the actual kernel degrees,

$$|G(B,U)| < 2C(1 + 3b + 9b^2)B^H \leq 26Cb^2B^H, \qquad 2G(B,U) < \frac{1}{64}B^{H+1}.$$

Define

$$T_{\mathrm{arg}} = \sum_j B^{q_j}, \qquad N_0 = B^{117}, \qquad N_1 = B^{2162}, \qquad J_{\mathrm{code}} = 2G(B,U) + T_{\mathrm{arg}}.$$

Since $Q = 2129$, both T_{arg} and $(B-2)T_{\mathrm{arg}}$ are less than $N_1/64$. Therefore

$$0 < J_{\mathrm{code}} < N_1/32, \qquad J_{\mathrm{code}} + (B-2)T_{\mathrm{arg}} + 2 < N_1.$$

If the usual low words of S and T are each below N_0, the latter strict inequality proves $S+T < N = N_0N_1$ including their possible low-block carry. All these inequalities hold before binary recovery. The older dense padding $2G + B\sum_{h=0}^{H} B^h$ also lies between 0 and $2B^{H+1}$, if an adapter retains $N_1 = 4B^{H+1}$.

After digit recovery write $g = \sum_i n_i B^{w_i}$ with $0 \leq n_i < b$ and $p_i = n_i + 1$. Then $U = \sum_i p_i B^{w_i}$ and $\sum_i p_i \leq 31b$. Every coefficient of the complete polynomial G is bounded in absolute value by

$$C\left((\sum_i p_i)^2 + \sum_i p_i + 1\right) \leq 993Cb^2 < B/4. \tag{5}$$

This controls unsupported collisions too; it sums the three homogeneous layers rather than assuming that all quadratic labels are distinct. The inherited sparse-borrow padding lemma now applies to $J_{\mathrm{code}} =$

$2G+T_{\arg}$. Its marked coefficients are exactly $2\rho_j$, by (2), and the row inverse recovers the original source equations.

Conversely, any positive source solution supplies $n_i = p_i - 1$ for every sufficiently large base. The all-one tuple is not a solution because the unchanged row 5 equals 1 there. Hence some $n_i > 0$, and the positive labels give $g \geq B > b$. Thus the lower source-witness bound is preserved, not merely the upper bound.

4.4 One-cell positivity repair for signed adapters

An adapter that must prove positivity for every integer g, rather than only $g \geq 0$, may use

$$G_+(B,U) = G(B,U) + B^{2162}.$$

This changes no target coefficient since $2162 > Q$, and increases the exact height only to 2162. Completing the square, and using the bounds above,

$$\begin{aligned} G_+(B,U) &\geq B^{2162} - |D_0(B)| - \frac{D_1(B)^2}{4D_2(B)} \\ &> B^{2162} - 2CB^{1719} - 2C^2B^{2125} > 0. \end{aligned}$$

Indeed the normalized subtractions $2C/B^{443}$ and $2C^2/B^{37}$ are each less than $1/4$ for $B \geq 4096C$. Positivity thus holds for every real U. The added off-target coefficient is 1, within the same coefficient bound; a sparse bounded-input adapter can take $N_1 = B^{2163}$.

4.5 What is and is not certified

The exact standard-library replay and ten corruption controls are in `verify_split.py` and `test_split.py`; ordinary and optimized Python produce byte-identical verification receipts. Neither search code nor a third-party symbolic package is a replay dependency. Compared with the frozen wave7 source, the convolution height falls from 5556 to 2161 (about 61.1%) and $H + M + 2$ falls from 5741 to 2279 (about 60.3%). These are source exponents, not final equation degrees. The finalizer must still be composed and its actual arithmetic circuit independently audited. No global optimum or new end-to-end proof-assistant certificate is claimed. The unbounded compiler/refutation and arithmetic finalizer theorems remain identified inherited semantic dependencies.

5 Degree-separated coding: a ten-integer equation

There is a completely fixed integer polynomial

$$F_{10}(f,g,h,k,l,w,x,y,z,m)$$

of exact total degree 8 469 528, evaluated by a division-free circuit of 533 binary arithmetic operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is a new source-coding architecture, not a new unknown-count theorem. The inherited compiler calibration and Sun arithmetic bridge remain stated mathematical dependencies. No consistency decision, global minimality claim, or new end-to-end proof-assistant certificate is asserted.

The structural change is to split the extraction convolution by homogeneous degree. A linear label may now collide with a quadratic label: their kernels are separate summands. This relaxes the finite isolation problem rather than merely finding a better placement for the previous problem. Elementary row operations, one shared positivity shift, borrow-tolerant target padding, and exact mask-population calibration complete the construction.

5.1 Fixed source and three homogeneous kernels

Use exactly the unshifted, strictly positive, 31-coordinate parent in `data/parent_quartic31.json`, with its complete original four-port binding $(\beta_{\mathrm{native}}, K_0, M_C, M_F)$. The four included compressed payloads have bit lengths $(1064921, 1070004, 1064921, 1064922)$; the last is `MF_source`, already shifted, and is not shifted again. `INSTANCE.json` is the active binding. No package-52 transport row or coefficient is spliced into this tuple. The pinned parent theorem is

$$\exists p \in \mathbb{Z}_{>0}^{31}\ r_1(p) = \cdots = r_{22}(p) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

Replace only r_{14} by $\rho_{14} = r_{14} + r_2$ and r_{15} by $\rho_{15} = r_{15} + r_3$; otherwise put $\rho_j = r_j$. The inverse row operations preserve the common zero set over every coefficient tuple. There are now 89 records: seven constant, 39 linear, and 43 quadratic.

In the parent coordinate order

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta),$$

choose the positive, distinct weights

$$(w_1, \ldots, w_{31}) = (1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The row order and targets are

$$(2, 22, 8, 15, 9, 11, 3, 17, 13, 18, 12, 20, 4, 21, 19, 1, 6, 10, 16, 14, 7, 5),$$
$$(28, 85, 286, 384, 593, 602, 642, 731, 906, 966, 1083, 1212, 1231, 1393,\\ 1461, 1471, 1678, 1719, 1732, 1880, 2107, 2129).$$

Let q_j denote the target assigned to row j, and write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For $d = |\alpha|$ put

$$e(\alpha) = \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0, 1, \\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2. \end{cases}$$

Define the fixed integer polynomials

$$D_d(Z) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} Z^{q_j - e(\alpha)}, \qquad U_p(Z) = \sum_i p_i Z^{w_i}, \qquad G_p(Z) = D_2(Z) U_p(Z)^2 + D_1(Z) U_p(Z) + D_0(Z).$$

All reversed exponents are nonnegative. The exact finite certificate checks, for every used degree-d monomial α and every degree-d monomial γ in 31 variables,

$$q_j - e(\alpha) + e(\gamma) = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma.$$

There are respectively $1, 31, 496$ expansion monomials in degrees $0, 1, 2$, and 22,544 source/expansion interactions. There is no requirement to separate a degree-one label from a degree-two label. Multinomial expansion and the displayed coefficient weights therefore give the universal identities

$$[Z^{q_j}] G_p(Z) = 2\rho_j(p) \quad (1 \le j \le 22). \tag{6}$$

The independent checker compares the entire extracted polynomials in all 31 coordinates and four ports, not just evaluations at test points.

Write $M = 116$, $Q = 2129$. The exact kernel degrees and convolution height are

$$\deg D_0 = 1719, \qquad \deg D_1 = 2027, \qquad \deg D_2 = 1929, \qquad H = \max_d(\deg D_d + dM) = 2161.$$

The leading term of D_2 is exactly $+Z^{1929}$, arising from EX in row five. Every kernel separately has distinct exponents. Replacing each port monomial in a coefficient by its strict bit bound and taking absolute integer coefficients gives a maximum majorant of bit length 2134927. Consequently, with

$$C_0 = 2^{2134927}, \qquad \kappa = 2134940, \qquad \beta_S = 2^\kappa,$$

every kernel coefficient has absolute value less than C_0, and $\beta_S > 4096C_0$.

5.2 Source polynomials and pre-binary bounds

In the unrestricted integer unknowns f, g define

$$b = 1 + 3f, \quad B = \beta_S b^2, \quad \ell = \sum_i B^{w_i}, \quad U_c = \ell + g,$$

$$G_c = D_2(B)U_c^2 + D_1(B)U_c + D_0(B), \quad A_T = \sum_j B^{q_j},$$

$$N_0 = B^{117}, \quad N_1 = B^{2162}, \quad N = N_0N_1 = B^{2279},$$

$$M_0 = N_0 - 1 - (b-1)\ell, \quad J_c = 2G_c + A_T,$$

$$S_c = g + J_cN_0, \quad T_c = M_0 + (B-2)A_TN_0,$$

$$X = N^2 + (S_c + T_c)N + N - 1 - S_c,$$

$$Y = \beta_S^{15}\, 2^{\kappa-21}B^{2402}b = 2^{2418\kappa-21}b^{4805}, \qquad \mathcal{C} = bB^{116} = 2^{116\kappa}b^{233}.$$

All powers have fixed nonnegative integer exponents. Every displayed register is an abbreviation, never an additional quantified unknown.

Assume $f \neq 0$, b is a square, and $0 \leq g < 2\mathcal{C}$. Then $b \equiv 1 \pmod 3$ implies $b \geq 4$, and $B > 4096C_0b^2$. Distinct positive weights give $\ell < 2B^{116}$, hence

$$B^{116} \leq U_c < 3bB^{116}.$$

In particular, without assuming b is a power of two,

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2C_0B^{2027}, \quad |D_0(B)| < 2C_0B^{1719}.$$

After division by $B^{1929}U_c^2$, the two possible negative terms are bounded by $2C_0/B^{18} < 1/8$ and $2C_0/B^{442} < 1/8$. Thus $G_c > \frac{1}{4}B^{1929}U_c^2 > 0$. This is the necessary signed leading-layer argument; positivity is not inferred from positive inputs alone.

The kernel degree bounds also give

$$|G_c| < 2C_0(1 + 3b + 9b^2)B^H \leq 26C_0b^2B^H, \qquad 2G_c < N_1/64.$$

Since $Q = 2129 < H$, both A_T and $(B-2)A_T$ are less than $N_1/64$. It follows that

$$0 < J_c < N_1/32, \qquad J_c + (B-2)A_T + 2 < N_1.$$

The digits of M_0 are $B-1$, except that the 31 designated cells have digit $B-b$. Therefore $0 \leq M_0 < N_0$, and $g < N_0$ as well. The preceding strict high-block bound proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{7}$$

Consequently $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. Because κ is even, $B \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$, and $T_c \equiv -22$; hence $X \equiv 1 + T_c \equiv 0 \pmod 3$. These are all established before binary-base recovery.

5.3 Borrow-tolerant extraction and exact mask population

Suppose now that b is a power of two. So are B, N_0, N. The low mask condition $g \,\&\, M_0 = 0$ is exactly

$$g = \sum_i n_i B^{w_i}, \quad 0 \le n_i < b.$$

For these digits put $p_i = n_i + 1$, so $U_c = \sum_i p_i B^{w_i}$ encodes positive parent coordinates without expanding a successor substitution in every residual. Write $h_s = [Z^s]G_p(Z)$ and let ϵ_s be the indicator of the 22 targets. All coefficients, including unsupported collisions, satisfy

$$|h_s| < C_0\{(\sum_i p_i)^2 + \sum_i p_i + 1\} \le 993 C_0 b^2 < B/4.$$

The signed expansion of J_c is $\sum_{s=0}^{H}(2h_s + \epsilon_s)B^s$. Euclidean digit normalization has incoming carries only in $\{-1, 0\}$:

$$u_0 = 0, \quad u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B} \right\rfloor, \quad d_s = 2h_s + \epsilon_s + u_s - B u_{s+1}.$$

At a marked cell the mask $B - 2$ admits precisely digits $0, 1$. If $h_s = 0$, the unnormalized digit $1 + u_s$ passes. If $h_s \ge 1$, it lies in $[2, B)$ and fails. If $h_s \le -1$, it lies strictly between $-B + 1$ and zero, and its normalized digit is at least two, so it fails. Thus incoming borrows do not affect the zero test. Positivity of J_c was proved independently, ensuring these are its ordinary digits. Combining the disjoint low and high blocks with (6) proves

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i},\ 0 \le n_i < b, \quad r_1(n+1) = \cdots = r_{22}(n+1) = 0.$$

For $N = 2^u$ and $S, T \ge 0$ with $S + T < N$, let $X_* = N^2 + (S+T)N + N - 1 - S$. Its disjoint binary blocks give

$$s_2(X_*) = 1 + u + s_2(T) - \{s_2(S) + s_2(T) - s_2(S+T)\}.$$

The braces count carries with multiplicity, and vanish precisely when $S \,\&\, T = 0$. Legendre's formula therefore implies

$$2^{1+u+s_2(T)} \mid \binom{2X_*}{X_*} \quad \Longleftrightarrow \quad S \,\&\, T = 0.$$

The stronger range (7) is essential. Writing $b = 2^k$, $B = 2^v$, the actual mask population is

$$s_2(T_c) = (117 + 22)v - 31k - 22, \qquad u = 2279v.$$

Hence $1 + u + s_2(T_c) = 2418v - 31k - 21 = \log_2 Y$, exactly explaining the fixed polynomial threshold used above.

Conversely, a positive parent zero is not the all-one vector: original row five equals one there. For arbitrarily large $b = 4^s$ exceeding all its coordinates, encode $n_i = p_i - 1$. Then $g \ge B > b$ and $g < \mathcal{C}$; the latter follows from $(b-1)\ell < bB^{116}$. All marked tests pass and $Y \mid \binom{2X}{X}$. This proves both directions of the entire source contract needed by the inherited finalizer, including its lower bound on forward witnesses and its pre-binary reverse bounds.

5.4 Explicit radical-free finalizer

The following formulas are the inherited package-31 quintic, merged-square, structural-weight finalizer. They are written out to fix the actual equation. Distinguish its local B_*, S_*, T_* from the source base and masks above. Define

$$
\begin{gathered}
L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw, \\
A = U(V+1), \qquad B_* = 2X+1, \qquad C = B_* + (A-2)h, \\
P = U^2V, \qquad K = X + 1 + k(P-2), \\
S_* = 2A - 5, \qquad T_* = 3WC - 2(W^2-1), \\
R_2 = (P^2-4)K^2 + 4, \\
\Delta = 8\mathcal{C}^3 g K^2 - g^2\{32(C-KL)^2\mathcal{C}^3 + g^2K^2\}, \\
O = f^2 l^2 \Delta, \qquad \xi = zx, \\
D = (A^2-4)C^2 + 4, \qquad E = C^2 D\xi, \\
F = 4(A^2-4)E^2 + 1, \qquad G_* = 2(A+2)(A-2)^2E^2 - 1, \\
\Omega = 16E, \qquad H_* = C + (B_* - C)F + \Omega F y, \\
I_* = (G_*^2 - 1)H_*^2 + 1.
\end{gathered}
$$

Put

$$
\begin{aligned}
r &= T_* + S_* y, & \Sigma &= \xi S_*^5, & & \\
\Theta &= (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2), & & & & \\
a_0 &= b\{1 + (3O-4)b^2z^2\}, & a_1 &= R_2, & a_2 &= DFI_*, \\
w_0 &= 1, & w_1 &= a_0, & w_2 &= a_0(P^2 + K^2 + 3), \\
Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 w_i^2 a_i \quad (0 \le i \le 2), & & \\
V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2A_2 - 4A_0A_1. & &
\end{aligned}
$$

The promised single equation is

$$
\boxed{F_{10} = W_n^2 - 16V_n^2Z^2A_2 = 0.} \tag{8}
$$

Every expression is an integer polynomial. In particular, radicals and binomial coefficients appear in the proof only, never in the equation.

Exact dependency boundary. Sun's 2021 Theorems 4.12–4.13, specialized as in his 2026 Propositions 6.1–6.2, provide the two bridge directions for any source satisfying the contract just proved. Forward witnesses have b square, R_2 square, $S_* \mid T_*$, $C = u_{B_*}(A)$ and $O > 0$. Reverse witnesses with b square, $O > 0$, $\xi \neq 0$, R_2, DFI_* square and $S_* \mid T_*$ recover $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$ and the required binomial divisibility. The residue interpolation theorem permits ξ to be divisible by any prescribed positive integer and permits any residue class for $y \pmod{\xi}$. The only data substituted here are the newly proved source polynomials; the bridge does not require $Y = N^2$.

For completeness, the inherited finalizer implications can be verified as follows. The polynomial $(v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and none over $\mathbb{Q}$. For odd primes other than 23 the cubic discriminant is -23: if the cubic is irreducible its Frobenius permutation is even, so the quadratic has a simple root; otherwise the cubic has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$. Hensel lifting and the Chinese remainder theorem finish the local-root argument. Prime valuations in the two homogeneous factors show that $\Sigma \mid \Theta$ forces $S_* \mid T_*$: if $v_p(S_*) > v_p(T_*)$, the

valuation of Θ is $5v_p(T_*) < 5v_p(S_*)$. Since S_* is odd, it never vanishes; the absence of rational roots ensures $\Theta \neq 0$.

When $z \neq 0$, a_0 square implies that its coprime factors b and $1 + (3O - 4)b^2z^2$ are signed squares. The congruence $b \equiv 1 \pmod 3$ excludes a negative square and excludes zero, including the zero-factor exception. Thus b is a positive square and the usual positivity Pell lemma gives $O > 0$. Conversely, forward witnesses have $O \geq 4$: the positive integer Δ is divisible by $g \geq b \geq 4$. For $d = 3O - 4 > 0$, $d \equiv 2 \pmod 3$ is nonsquare. A positive Pell unit raised to its order modulo b gives $v = bz > 0$ with $u^2 - dv^2 = 1$, making a_0 a square. Residue interpolation then supplies $\xi = zx$ and a y that simultaneously makes a_2 square and $\Sigma \mid \Theta$.

If $z \neq 0$, the displayed a_0 always has $|a_0| > 1$. The exceptional possibilities $a_0 = 0, \pm 1$ force $|b| = 1$, hence $b = 1$, $f = O = 0$, and $a_0 = 1 - 4z^2$, a contradiction. Also $\sqrt{|a_1|} \leq P^2 + K^2 + 2$. Thus the weights obey

$$|w_1| > \sqrt{|a_0|}, \qquad |w_2| > \sqrt{|a_0|} + |w_1|\sqrt{|a_1|}.$$

In the multiquadratic field, an automorphism changing a nonzero weighted root has a largest changed index; its magnitude exceeds the sum of all smaller changed magnitudes. A rational weighted sum can therefore occur only when every a_i is an integer square. This also covers negative radicands by complex absolute values, and zero radicands separately.

The exact norm identity

$$F_{10} = \prod_{\epsilon \in \{\pm 1\}^3} \left(Z + \sum_{i=0}^{2} \epsilon_i \sqrt{A_i} \right)$$

has 35 expanded terms and is checked independently by integer polynomial arithmetic. If $\Sigma = 0$ then $F_{10} = \Theta^8 \neq 0$. At a zero of F_{10} we therefore have $\Sigma \neq 0$, hence $z \neq 0$; separation forces all a_i square and $\Sigma \mid \Theta$. The reverse bridge and source proof recover a parent solution. Conversely the forward construction just described makes all a_i square and $\Sigma \mid \Theta$; choosing m to cancel one signed factor gives $F_{10} = 0$. Composition with the pinned parent theorem proves the stated consistency equivalence in ZFC.

5.5 Exact circuit, degree, and reproducibility

The actual circuit uses the same finalizer polynomial in factored form. For example, with $A = 2A_{1/2}$ and $F - 1 = 4(A^2 - 4)E^2$, its Lucas expression is $G_* = (A_{1/2} - 1)(F - 1) - 1$; also $H_* = C + F(B_* - C + \Omega y)$. The common factor $\Sigma^2 a_0^2$ is reused in the last two scaled radicands. Fixed powers reuse already computed powers B^i, B^j when $i+j$ is the required exponent, falling back to binary exponentiation otherwise. These are exact polynomial identities, not a relaxation of the equation.

`generated/sun10.json` contains all ten variable leaves and 533 live binary gates: 340 multiplications, 166 additions, and 27 subtractions. Multiplication by a numeral is charged. The fixed coefficient leaves are specified by `INSTANCE.json`, the four payloads, and `generated/source_records.json`. Their construction is not included in this arithmetic-operation count, which is not bit complexity.

The source degrees are $\deg X = 9116$, $\deg Y = 4805$, and $\deg \mathcal{C} = 233$. The propagated final upper bound is

$$456(9116) + 896(4805) + 24(233) + 1760 = 8\,469\,528.$$

After all ten variables are replaced by one indeterminate T, the coefficient of $T^{8469528}$ has nonzero residues

p	101	1009	1000003
$[T^{8469528}]F_{10}(T, \ldots, T) \bmod p$	68	256	476348.

Any one residue proves the matching lower bound, so this degree is exact. This is leading-coefficient propagation, not a random point evaluation.

The independent checker imports no builder. It independently parses all 22 source residuals; verifies the inverse row operations, 89 records, and all 22,544 isolation interactions; checks 59 local circuit

identities, the additional Lucas factorization, and the eight-factor norm identity; audits every gate and fixed binding; and certifies the exact degree. Finite implementation tests cover 476 borrow transitions, 11,049 binary block pairs, and 60 large-integer source assignments, including nonbinary square bases 25 and 49. These finite checks supplement, rather than replace, the unbounded mathematical arguments. Both ordinary Python and `python3 -O independent_check.py` pass.

Against the prior calibrated-mask ten-integer construction of degree 21 111 400 and 659 gates, the new point has 59.88% lower degree and 126 fewer gates (19.12%). Against the incoming 582-gate homogeneous source point of degree 50 822 936, it has 49 fewer gates and more than sixfold lower degree. These are explicit pinned comparisons, not claims of global optimality or superiority over every variable/domain tradeoff.

The inherited exact-source pin is `124c07c0b77a`; `SOURCE_PINS.json` preserves all 12 source identities. Primary arithmetic dependencies are Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Sci. China Math. 64 (2021), 281–306, https://arxiv.org/html/1704.03504v7; and *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, https://arxiv.org/html/2609.01594v2. The complete inherited source, coefficient, and finalizer proof records are preserved under `upstream/`; their historical relative paths do not supersede the active manifest.

6 A degree-separated nine-variable equation

The primary equation has nine strictly positive unknowns, exact total degree 2 676 548, and 476 live binary operations. Its separately translated nonnegative form has the same degree. The complete division-free DAGs, coefficient payloads, exact-degree certificates, and a standard-library checker accompany this section. Register names below abbreviate polynomials; they introduce no additional unknowns.

Scope. The ordinary mathematical theorem is that ZFC proves existence of a zero in the stated domain equivalent to existence of a finite syntactic ZFC refutation. Its negation is the requested consistency equivalence. The original package31 compiler and the numerical Jones ratio theorems are explicit imported dependencies. This is not a proof of consistency, a new end-to-end kernel formalization, an unrestricted-integer nine-variable claim, or a global minimum claim.

6.1 A smaller source by keeping the three degrees separate

Use the original 31-positive-variable, 22-residual source $r_1, \dots, r_{22}$, with its *whole* original coefficient tuple. The four payload bit lengths are $1064921, 1070004, 1064921, 1064922$; the fourth port is already shifted. Replace just

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3.$$

The inverse row operations recover the old system, so its zero set is unchanged. It has 89 weighted records: 7 constant, 39 linear, and 43 quadratic.

For positive source coordinates y_i, write $U_y(t) = \sum_i y_i t^{w_i}$, with weights

$$(1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The maximum weight is $W = 116$. For a monomial of degree d, weight its coefficient by 2 when $d = 0, 1$, and by $\prod_i \alpha_i!$ when $d = 2$. With the fixed target assigned to its row, form

$$D_d(t) = \sum_{j, |\alpha| = d} c_{j,\alpha} t^{q_j - \sum_i \alpha_i w_i}, \qquad \mathcal{G}(t, y) = D_2(t) U_y(t)^2 + D_1(t) U_y(t) + D_0(t).$$

The finite target certificate makes a degree-d term hit a marked target only against its own monomial of that same degree. Cross-degree label collisions are harmless, because the three kernels are separate summands. Checking all 22 544 interactions proves the formal identities

$$[t^{q_j}]\mathcal{G}(t,y) = 2\rho_j(y).$$

The kernel degrees are $(1719, 2027, 1929)$, the leading coefficient of D_2 is $+1$, and the convolution height is $K_{\mathrm{code}} = 2161$. The largest target is 2129. All 89 kernel records, coefficient recipes, and the row-target map are supplied as exact JSON.

6.2 Positive words and unconditional source bounds

Every weighted coefficient has absolute value below $M = 2^{2134927}$. Set the fixed ordinary integer coefficients

$$a = 4096M = 2^{2134939}, \qquad c = 2^{17079502}.$$

Use mixed coordinates $b, g, h, s, w, i \geq 1$ and $\phi, j, n \geq 0$. Define

$$\begin{aligned}
&B = ab^2, && \ell = \sum_i B^{w_i}, && U_s = \ell + g,\\
&G_s = (D_2(B)U_s + D_1(B))U_s + D_0(B), && A_q = \sum_j B^{q_j}, &&\\
&N_0 = B^{117}, && Z = B^{2162}, && M_c = N_0 - 1 - (b-1)\ell,\\
&J_c = 2G_s + A_q, && S_c = g + N_0 J_c, && T_c = M_c + N_0(B-2)A_q,\\
&L = 2bB^{2279}, && N = cbB^{1201}, && G_{\mathrm{cap}} = 32bB^{116},\\
&R = L(L + S_c + T_c + 1) - S_c - 1. &&&&
\end{aligned}$$

Here L is the binary-block length, while N is the separate square-root threshold for Jones's theorem.

Positivity must precede digit recovery. For every $g \geq 1$, $U_s \geq B^{116}$, and geometric bounds give

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2MB^{2027}, \quad |D_0(B)| < 2MB^{1719}.$$

Relative to $B^{1929}U_s^2$, the negative contributions are bounded by $2M/B^{18}$ and $2M/B^{442}$, each below $1/8$. Thus $G_s > \frac{1}{4}B^{1929}U_s^2 > 0$. Also $0 \leq M_c < N_0$, and

$$R = L^2 + (L-1)S_c + LT_c + L - 1 \geq L^2 + L - 1.$$

Consequently $R, N \geq 8$, $b \leq R, N$, and $G_{\mathrm{cap}} > 0$ on every outer assignment, without first assuming the final equation vanishes.

On the branch $g < 4bB^{116}$, one has $U_s < 6bB^{116}$, and

$$|G_s| < \frac{43Mb^2B^{2162}}{B-1}, \qquad 2G_s < Z/32, \qquad A_q, (B-2)A_q < Z/64.$$

Hence $g < N_0$, $J_c < 3Z/64$, and

$$S_c + T_c < N_0 Z = B^{2279} < L.$$

These are numerical, prebinary bounds, not circular consequences of the carry test.

If b is a power of two, so is B. The low-mask test $g \,\&\, M_c = 0$ says exactly $g = \sum_i z_i B^{w_i}$, $0 \leq z_i < b$. Then $y_i = z_i + 1$ are positive and $U_s = \sum_i y_i B^{w_i}$. Each coefficient h_t of $\mathcal{G}(t,y)$, including unsupported collisions, satisfies

$$|h_t| < M\big((\textstyle\sum_i y_i)^2 + \sum_i y_i + 1\big) \leq 993Mb^2 < B/4.$$

The signed digits of J_c are $2h_t+1$ at targets and $2h_t$ elsewhere. Normalization has incoming and outgoing borrows in $\{-1,0\}$. At a target, the normalized digit avoids $B-2$ exactly when $h_t=0$: the allowed digits are 0 or 1, while positive or negative h_t activate a masked bit. Therefore

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive common zero of the source.}$$

Conversely, any source zero admits every sufficiently large power-of-two b; its digits are $z_i = y_i - 1$. The all-one source tuple is not a zero, because $r_5(1,\dots,1) = 1$. Thus $g > 0$, and $g < bB^{116}$, giving the stronger completeness bound $32g < G_{\mathrm{cap}}$.

6.3 A calibrated two-block threshold

For $S, T \geq 0$, $S + T < L = 2^u$, the integer

$$X = L^2 + (S+T)L + (L-1-S)$$

has nonoverlapping binary blocks. Consequently

$$\mathrm{pop}(X) = u + \mathrm{pop}(T) + 1 - \big[\mathrm{pop}(S) + \mathrm{pop}(T) - \mathrm{pop}(S+T)\big].$$

The bracket is nonnegative and vanishes exactly for carry-free addition. Legendre's formula identifies $\mathrm{pop}(X)$ with $v_2\binom{2X}{X}$.

Write $b = 2^k$, $B = 2^v$, $v = 2134939 + 2k$. The unchanged low mask has $117v - 31k$ set bits and the 22 target cells contribute $22(v-1)$. Hence

$$\mathrm{pop}(T_c) = 139v - 31k - 22, \qquad u = 1 + k + 2279v.$$

The exact threshold is

$$u + \mathrm{pop}(T_c) + 1 = 2418v - 30k - 20 = 2(17079502 + k + 1201v) = 2\log_2 N.$$

The factor b in N accommodates the even parameter 2418, without restricting b to a sparse family. We have proved

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

6.4 Ratio registers and the positivity-first finalizer

Define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & &\\
D &= (A^2-1)C^2 + 1, & k_0 &= iJC, & E &= k_0 DC,\\
F &= (A^2-1)E^2 + 1, & G &= A + F(CD + 1 - A), & &\\
H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1, & &\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI. & &
\end{aligned}$$

Since $F \equiv 1 \pmod J$ and $F \equiv 1 \pmod C$, the old divisibility $JF \mid QF + J(H_{\mathrm{old}} - C)$ is equivalent to $J \mid Q$ and $F \mid H_{\mathrm{old}} - C$. The exact witness substitution

$$H = C + F(B_1 + jC), \qquad j_{\mathrm{old}} = 1 + \frac{F-1}{C} B_1 + Fj \geq 1$$

parametrizes precisely those old witnesses; the quotient is an integer. It introduces no new variable.

Use the polynomial margin with its known factor 8 removed:

$$t = bB^{116}, \qquad m = K^2(4t - g) - 16t(C - KY)^2.$$

Then $8m$ is exactly the original Jones margin. Its strict sign is unchanged. The imported ratio interface is now

$$a_0 \text{ square}, \quad a_1 \text{ square}, \quad J \mid Q, \quad m > 0.$$

Soundness gives b a power of two, $8g < G_{\mathrm{cap}}$, and the central-binomial divisibility. Completeness uses $32g < G_{\mathrm{cap}}$. All of its numerical hypotheses were established before this invocation.

The strict positive majorants

$$V_0 = PK, \qquad T_* = k_0(8D^2 - 4D - 1), \qquad V_1 = T_*GH$$

satisfy

$$\begin{aligned} V_0^2 - a_0 &= K^2 - 1 > 0, \\ T_*^2 - 64DF &= (8D+1)k_0^2 - 64D > 0, \\ \Delta_1 := V_1^2 - 64a_1 &= \big((8D+1)k_0^2 - 64D\big)(GH)^2 + 64DF(H^2-1) > 0. \end{aligned}$$

Also $0 < Q/J < Q < 3C^2 < 3D < T_* \le V_1$. The promised equation is

$$\Omega = Jm, \qquad \Theta = Jn + Q - \Omega(V_0 + V_1).$$

$$\boxed{F_9 = \left[\Theta^2 - \Omega^2(a_0 + 64a_1)\right]^2 - 256\Omega^4 a_0 a_1 = 0.}$$

If $m = 0$, then $\Theta = Jn + Q > 0$, so there is no zero. Otherwise the real norm factors give roots $r_0^2 = a_0$, $r_1^2 = 64a_1$ and

$$n + Q/J = m(V_0 + V_1 + r_0 + r_1).$$

The majorants make the parenthesis positive before any integrality claim. Thus $m > 0$. The margin gives $C > K$, and the elementary inequalities $A^2 > P$, $D > C^2$, $F \ge A^2$, $G \ge A$, $I \ge G^2$ give $a_1 > a_0$. Hence the rational signed sum $x = \Theta/\Omega = r_0 + r_1$ is nonzero. From $2xr_0 = x^2 + a_0 - 64a_1$, both roots are rational and therefore integral; $r_1/8$ is integral as well. The factor equation then yields $J \mid Q$. Conversely choose positive roots and set $n = m(V_0 + V_1 + r_0 + r_1) - Q/J > 0$. This proves the finalizer equivalence.

6.5 Exact degree and implementation certificate

The diagonal degrees are $r = \deg R = 9118$, $t = \deg N = 2403$. The degree-certificate circuit resolves the near-square cancellation by

$$\begin{aligned} &\Theta^2 - \Omega^2(a_0 + 64a_1) \\ &\quad = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\big[(K^2 - 1) + \Delta_1 + 2V_0V_1\big], \qquad E_0 = Jn + Q. \end{aligned}$$

The unique dominant term is $\Omega^2\Delta_1$, of degree 1,338,274. Its square dominates the final subtracted norm term, of degree 1,823,794. Thus

$$\deg F_9 = 188r + 400t + 8W + 236 = 2676548, \qquad \mathrm{lc}_{\mathrm{diag}}(F_9) = 2^{542}a^{1337768}c^{400} > 0.$$

The diagonal attains the full multivariate upper bound, proving equality. The positive coordinates replace ϕ, j, n by their positive inputs minus 1; the natural version instead increments the other six inputs. These are domain bijections, not integer-square parametrizations.

The supplied checker authenticates both separate whole tuples, rebuilds all 16 primary/comparison and certificate circuits, proves 24 source-evaluator polynomial identities, and replays 480 actual-tuple modular assignments. It additionally exercises 43,945 binary-block cases, 540 threshold identities, Pell-majorant and CRT identities, and finalizer completeness. These finite checks audit the implementation; they do not replace the universal proof. The source and full theorem are detailed in `nine/PROOF.md`.

7 Thirteen integers: CRT elimination and a coprime spectator guard

The polynomial specified below has exactly thirteen unrestricted integer unknowns, exact total degree 11 546 112, and a division-free circuit of 520 live binary operations (328 multiplications, 169 additions and 23 subtractions). Its integer zero existence is equivalent to a finite syntactic ZFC refutation. Thus, with the stated inherited compiler theorem, ZFC proves that its lack of integer zeros is equivalent to Con(ZFC). This is an ordinary mathematical proof, not a new Lean certificate, a proof of consistency, or a global minimality claim. Unlike the ten-integer companion, the arithmetic finalizer here uses only the established Jones ratio interface, not Sun's 2026 bridge. The recommended compact companion in Section 7.8 has degree 11 543 632 and 518 operations, with the same thirteen unknowns.

7.1 Degree-separated kernels and the complete polynomial

Use the original package-31 system of 22 quadratic residuals $r_j(p)$ in 31 strictly positive coordinates. The exact data are `split/parent_source.json`; the coherent fixed port tuple is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING.json`. All four compressed little-endian payloads are included and hash-authenticated. The inherited compiler theorem identifies its positive common zeros with finite syntactic ZFC refutations. This is the original tuple, not a mixture with a later packet's smaller coefficients.

Perform just two elementary integer row operations:

$$\varrho_{14} = r_{14} + r_2, \qquad \varrho_{15} = r_{15} + r_3, \qquad \varrho_j = r_j \quad (j \neq 14, 15).$$

The inverses are subtraction of rows 2 and 3, so the common zero set is unchanged. Write $\varrho_j = \sum_\alpha r_{j,\alpha} p^\alpha$ and

$$c_{j,\alpha} = \begin{cases} 2r_{j,\alpha}, & |\alpha| = 0, 1, \\ (\prod_i \alpha_i!) r_{j,\alpha}, & |\alpha| = 2. \end{cases}$$

Use the 31 positive weights w_i and the 22 targets q_j in `split/split_code.json`, and set

$$D_d(T) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} T^{q_j - \alpha\cdot w} \quad (d = 0, 1, 2).$$

The supplied exact certificate establishes

$$W = \max_i w_i = 116, \quad Q_* = \max_j q_j = 2129, \quad (\deg D_0, \deg D_1, \deg D_2) = (1719, 2027, 1929).$$

The leading coefficient of D_2 is exactly 1; every kernel has internally distinct exponents. There are 89 source records. Checking all 22,544 same-degree source/expansion interactions proves

$$[T^{q_j}]\big(D_2(T)U(T)^2 + D_1(T)U(T) + D_0(T)\big) = 2\varrho_j(p), \qquad U(T) = \sum_i p_i T^{w_i}.$$

The degrees are kept in separate products. Thus a linear label may equal a quadratic label without causing cross-degree contamination. No homogeneous dummy coordinate or full Sidon condition is needed. Let

$$K_* = \max_d(\deg D_d + dW) = 2161, \qquad E_* = K_* + W + 2 = 2279.$$

The fixed coefficient bound, checked from the actual original port bit lengths, is $|c_{j,\alpha}| < A_* = 2^{2134927}$.

The unknowns are exactly

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

All following expressions are polynomial abbreviations. Fix

$$\kappa = 2134940, \qquad a = 2^{\kappa} = 8192A_*, \qquad \eta = 2^{17079510} = 2^{8\kappa-10},$$

and define

$$\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
U_c &= \ell + g, & \mathcal{G} &= D_2(B)U_c^2 + D_1(B)U_c + D_0(B),\\
\mathcal{G}_+ &= \mathcal{G} + B^{2162}, & N_0 &= B^{117},\\
L &= 2bB^{2279}, & N &= \eta bB^{1201},\\
c &= bB^{116}, & G_c &= 32c,\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2\mathcal{G}_+ + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c).
\end{aligned}$$

The extra monomial B^{2162} lies above every target. It repairs positivity on arbitrary signed g without changing an extracted residual or adding a new unknown. Every exponent in these formulas is fixed and nonnegative.

Use the Jones registers with the exact CRT reparameterization of H below, and $\omega = w^2$:

$$\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= J, & t &= Q.
\end{aligned}$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s - 1, 3\phi + 2, 3\Delta - 1).$$

The new coprime-spectator stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (j = 1, 2),$$
$$B_3^* = 1 + 4d_3C_2a_1z_3^2, \qquad C_* = C_3 = C_2B_3^*.$$

Thus the last guard contains the second square candidate a_1 as a spectator. This is the only change to the Pell sign stack. Finally define

$$\begin{aligned}
Z &= \epsilon wiz_1z_2z_3a_0, & \Sigma &= Zu^5, & \rho &= t + ur,\\
\Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta,\\
A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 a_1.
\end{aligned}$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{9}$$

The literal division-free DAG is `generated/signed_coprime_split_crt.json`.

7.2 Signed-input positivity and sharp block bounds

Assume only $\epsilon \neq 0$, hence $b \geq 2$ and $B = 8192A_*b^2$. Geometric-series estimates give

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2A_*B^{2027}, \quad |D_0(B)| < 2A_*B^{1719}.$$

Completing half of the quadratic square gives, for every real U_c,

$$\begin{aligned}\mathcal{G}_+ &\geq \tfrac{1}{2}D_2(B)U_c^2 + B^{2162} - \frac{D_1(B)^2}{2D_2(B)} - |D_0(B)| \\ &> \tfrac{1}{4}B^{1929}U_c^2 + B^{2162} - 4A_*^2B^{2125} - 2A_*B^{1719} \\ &> \tfrac{1}{4}B^{1929}U_c^2 + \tfrac{1}{2}B^{2162}.\end{aligned}$$

The last line uses $4A_*^2/B^{37} < 1/4$ and $2A_*/B^{443} < 1/4$. For every *integer* g, $U_c = \ell + g$ is an integer. Therefore

$$S_c > \tfrac{1}{2}N_0B^{1929}U_c^2 + U_c + N_0B^{2162} - \ell > 0.$$

Indeed $tU_c^2 + U_c \geq 0$ for an integer U_c and $t \geq 1$, while $\ell < N_0$. This explicitly absorbs the signed linear term g, rather than incorrectly deducing $S_c > 0$ merely from $J_c > 0$.

The mask satisfies $0 < M_c < N_0$, so $T_c > 0$ as well. Consequently

$$R = L^2 + S_c(L-1) + T_cL + L - 1 > L^2.$$

Also

$$N^2 = 2^{-20}a^{15}B^{2403} < B^{2418} < B^{4558} < L^2, \qquad b \leq c \leq N.$$

Thus $R \geq N \geq 8$, $b \leq R, N$, and all these estimates hold before any sign guard, carry condition, or source-coordinate decoding.

For the range needed by the interval guard, assume $0 < g < c$. Since $\ell < 2B^{116}$, one has $U_c < 2bB^{116}$. Then

$$|\mathcal{G}| < 12A_*b^2B^{2161}, \qquad 2|\mathcal{G}| < B^{2162}/256, \qquad T_* < B^{2162}/256.$$

Writing $P_c = B^{2162}$, this gives $0 < J_c < (2 + 1/128)P_c$. Since $g < N_0$ and $P_c > 128$,

$$0 < S_c < 3B^{2279}, \qquad 0 < T_c < B^{2279}, \qquad S_c + T_c < 4B^{2279} \leq L.$$

For the middle bound, use $(B-2)T_* < B^{2130}$ and $1 + B^{2130} < B^{2162}$. These sharp estimates are why the off-target padding does not require increasing the block exponent from 2279 to 2280.

7.3 Sparse carries and parity-tuned square threshold

Continue to assume $0 < g < c$, and suppose now $b = 2^k$, so $B = 2^\nu$ with $\nu = \kappa + 2k$. The low mask gives

$$g \,\&\, M_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_iB^{w_i}, \qquad 0 \leq n_i < b.$$

Here the binary "and" is proof notation only. Put $p_i = n_i + 1$, so $U_c = \sum_i p_iB^{w_i}$. Every coefficient h_e of $D_2(T)U(T)^2 + D_1(T)U(T) + D_0(T) + T^{2162}$ obeys

$$|h_e| < 994A_*b^2 < B/8.$$

The bound includes the possible collisions at unmarked coefficients and the extra coefficient 1; it follows from $\sum_i p_i \leq 31b$. The unnormalized digits of J_c are $2h_e + \varepsilon_e$, where $\varepsilon_e = 1$ exactly at the targets. Starting with incoming carry 0, normalized carries remain in $\{-1, 0\}$. At a target, its digit avoids $B - 2$

exactly when it is 0 or 1. This happens exactly when $h_e = 0$, independently of an incoming borrow. The degree-separated extraction theorem then gives, using the inverse row operations,

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad p = n + 1 \text{ is a positive common zero of the original system.}$$

Conversely every such zero has this code with $0 < g < c$ for arbitrarily large $b = 2^{1+2r}$. The all-one positive tuple is not a zero, since original row 5 has value 1 there. Thus the encoded nonnegative tuple is not zero.

Write $L = 2^\lambda$. Since $S_c, T_c > 0$ and $S_c + T_c < L$, the three nonoverlapping blocks in R give

$$s_2(R) = \lambda + s_2(T_c) + 1 - \tau_2(S_c, T_c),$$

where $\tau_2(S,T) = s_2(S) + s_2(T) - s_2(S+T) \geq 0$ is the number of binary carries. The 31 low allowed positions and 22 high target positions give

$$s_2(T_c) = 139\nu - 31k - 22, \qquad \lambda = 1 + k + 2279\nu.$$

Therefore

$$\begin{aligned} \lambda + s_2(T_c) + 1 &= 2418\kappa + 4806k - 20 \\ &= 2\big((8\kappa - 10) + k + 1201(\kappa + 2k)\big) = \log_2 N^2. \end{aligned}$$

Legendre's identity $v_2\binom{2R}{R} = s_2(R)$ now proves

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

The factor b in N is necessary in this parity class. It is a polynomial factor, not a new witness or a variable exponent.

7.4 An exact polynomial CRT reparameterization

The inherited ratio interface uses $H_{\text{old}} = B_1 + j_{\text{old}}C$ and $JF \mid QF + J(H_{\text{old}} - C)$. Since $E = iJDC^2$,

$$F \equiv 1 \pmod{J}, \qquad F \equiv 1 \pmod{C},$$

and hence $\gcd(J, F) = 1$. The divisibility is equivalent to $J \mid Q$ and $F \mid H_{\text{old}} - C$. Our changed register

$$H = C + F(B_1 + jC)$$

automatically satisfies the latter. It also has the original form, with

$$j_{\text{old}} = 1 + (A^2 - 1)i^2J^2D^2C^3B_1 + Fj,$$

which is a polynomial integer, since $F - 1 = (A^2 - 1)i^2J^2D^2C^4$. This identity holds for arbitrary signed inputs, including $C = 0$.

Conversely, at the canonical positive ratio witnesses, $0 < B_1 < C$ and $j_{\text{old}} \geq 1$. Therefore $z = (H_{\text{old}} - C)/F > 0$ and $z \equiv B_1 \pmod{C}$, so $z = B_1 + jC$ with $j \geq 0$. Thus every canonical witness is represented. No extra unknown or cofinal restriction is introduced. The merger now has $u = J$ rather than JF. This CRT step is inherited from the separately identified sparse-Pell packet, not claimed as an original theorem here.

7.5 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c-g) = c^2 - (c-2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK-1)^2 < (P^2-1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\mathrm{aux}} = 8gK^2 + 1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\mathrm{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ now means $J \mid Q$, while $F \mid H - C$ holds identically. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put $d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $J \mid Q$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U+1)^{2R}/U^R = Y + f$ with $0 \leq f < 2^{2R}/U = b/(2N^2)$ and $Y \geq 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \leq \frac{2(Y+f)}{Y(U+1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \geq 8$. Since $c, b \leq N$ and $g(c-g) \geq 1$, it follows that

$$\frac{c^2(C-KY)^2}{K^2} < \frac{c^2 b^2}{N^4} \leq 1 \leq g(c-g),$$

so the very same canonical witnesses, translated by the exact CRT formula, satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

7.6 Raw two-root recovery, including the zero spectator branch

For every integer assignment, $u = J = 4A - 5 \neq 0$. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either the cubic has a simple root or it is irreducible, whose Frobenius three-cycle makes its nonzero

discriminant -23 a square. The quadratic then has a simple root. At 23 the cubic has the simple root 3; at 2, $-23 \equiv 1 \pmod 8$ admits compatible square roots. Hensel lifting and CRT finish the assertion. Thus $\Theta = u^5\Pi(t/u + r) \neq 0$ for every assignment, and

$$\exists r \quad Zu^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

Indeed a root modulo $|Z|$ proves the reverse implication once $u \mid t$. If $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contrary to $u^5 \mid \Theta$. If $Z = 0$, divisibility would require $\Theta = 0$, already impossible.

At a zero of the equation, $\Sigma = 0$ is impossible because then the polynomial is $\Theta^4 \neq 0$. In particular $\epsilon, w, i, z_1, z_2, z_3, a_0$ are nonzero. Every $d_j \equiv 2 \pmod 3$ is nonzero, and every $B_j^* \equiv 1 \pmod 4$ is nonzero. Thus C_2, C_* are nonzero.

No coincident radicands. One has $C_* \neq a_1$ without any positivity assumption. If $a_1 = 0$ this follows from $C_* \neq 0$. Otherwise $B_3^* \neq 1$, and it cannot be -1 because it is 1 modulo 4. But

$$\gcd(B_3^*, a_1) = 1,$$

while $C_* = a_1$ would make B_3^* divide a_1. This would force $B_3^* = \pm 1$, a contradiction.

Put $q = X/\Sigma$. The raw norm equation implies

$$(q^2 + C_* - a_1)^2 = 4C_*q^2.$$

If $q = 0$, the norm would give $C_* = a_1$, just excluded. Therefore

$$r_0 = \frac{q^2 + C_* - a_1}{2q}, \qquad r_1 = q - r_0$$

are rational, with $r_0^2 = C_*$ and $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. In particular $J \mid Q$. No complex-root separation assumption is used.

Recover the first two guards before ruling out $a_1 = 0$. The factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime, because each new factor is 1 modulo its entire preceding product. Their nonzero product C_* is a square. Thus each absolute factor is a square. Since $B_j^* \equiv 1 \pmod 4$, every B_j^* is positive: a negative square cannot be 1 modulo 4. The product sign then forces $a_0 > 0$, also a square. All prefixes are now positive, so $d_1, d_2 > 0$, yielding $s \geq 1$ and $\phi \geq 0$.

At this point the last factor might still equal 1 if $a_1 = 0$; this branch must not be silently discarded. The unconditional source estimate above has already proved $R > L^2 > 0$ for *every* signed g. Since $\epsilon, w \neq 0$ and $s \geq 1$, we have $b \geq 2$, $N \geq 8$, $U, Y > 0$, $A \geq 2$, and, using $\phi \geq 0$, $C > 0$. Then

$$D \geq A^2 > 0, \qquad F \geq 1, \qquad CD + 1 - A > 0, \qquad G \geq A > 1, \qquad I \geq 1.$$

Consequently $a_1 = DFI > 0$. Now $B_3^* \neq 1$, and its positivity with $C_2 a_1 z_3^2 > 0$ forces $d_3 > 0$, hence $\Delta > 0$. This closes the zero-spectator branch without adding a_1 to the nonzero multiplier. The interval and ratio arguments above therefore apply.

Completeness. Start from the canonical positive ratio witnesses, translated by the CRT parameterization. They have a_0, a_1 positive squares, $J \mid Q$, and all three strict sign conditions. For $j = 1, 2$, $d_j C_{j-1}$ is positive nonsquare because $d_j \equiv 2 \pmod 3$ and C_{j-1} is a square. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even, so $z_j = y/2 \neq 0$ makes $B_j^* = x^2$. For the last guard, $d_3 C_2 a_1$ is likewise positive nonsquare, since $C_2 a_1$ is a positive square. The same construction gives B_3^* square and C_* square. The intersective lemma chooses r with $\Sigma \mid \Theta$. Finally choose

$$v = \Theta/\Sigma + \sqrt{C_*} + \sqrt{a_1} \in \mathbb{Z}.$$

Then the raw norm vanishes. Both directions are proved. By the inherited compiler theorem, integer-zero existence is equivalent to a finite ZFC refutation, and ZFC proves its negation equivalent to Con(ZFC).

7.7 Exact degree, comparison, and verification scope

The exact relevant register degrees are

	a_0	a_1	C_2	C_*	Σ	X
deg	261250	2635110	1045009	4875914	448571	448572

and $\deg A_0 = 5773056$, $\deg A_1 = 3532252$. The degree of A_0 strictly exceeds those of X^2 and A_1, so the output has leading term A_0^2. Its diagonal leading coefficient is

$$116361\, a^{2885464}\eta^{960} \pmod{1000003},$$

which is nonzero since a and η are powers of two. The circuit upper bound and this specialization prove exact degree 11 546 112.

The previous seeded thirteen-integer construction has degree 57 866 184 and 759 operations. At its unchanged old source, the coprime spectator alone gives degree 46 680 720 and 758 operations, and adding CRT gives degree 28 847 344 and 754 operations. With degree-separated extraction and the one-monomial signed-input repair, the canonical result is degree 11 546 112 and 520 operations: 80.05% less degree and 31.49% fewer operations than the seeded predecessor. All variants have thirteen unrestricted integer unknowns and the same original coherent coefficient tuple. The old-source proof is retained in `old_source_section.tex` for an explicit apples-to-apples comparison. No twelve-variable result, variable-count record, global optimum, or reduction of coefficient storage is claimed.

The metric counts every live binary addition, subtraction and multiplication, including scalar multiplication and all fixed-power gates. Reuse and fixed leaves are free. The canonical evaluator uses sparse Horner forms, reuses known exponent sums, and factors the merger cubic as $\rho(\rho^2 - u^2) - u^3$. These are exact polynomial identities, not new semantic assumptions.

The checker independently parses all 22 original residuals, proves the two row inverses and all 22 target polynomial identities, verifies the complete 22,544 isolation interactions, and authenticates the full fixed tuple. It also replays literal circuit registers over three finite fields, checks signed and zero branches, the norm and CRT identities, nonvacuous Pell examples, source positivity, sparse carries, and the parity calibration. These finite implementation audits corroborate the universal proof; they are not a new kernel-checked formal theorem or an unbounded consistency search.

Run `python build_jones.py`, `python check_jones.py`, and `python -O check_jones.py`. The complete division-free polynomial and fixed binding are in the emitted DAG and `FIXED_BINDING.json`. Original source provenance is in `SOURCE_PINS.json`; the CRT source archive and theorem are identified in `CRT_PROVENANCE.json`.

7.8 Recommended compact companion: odd-parity calibration

The recommended compact companion has the same thirteen unrestricted integer unknowns, exact degree 11 543 632, and 518 live binary operations (326 multiplications, 169 additions, 23 subtractions). Its literal DAG is `generated/signed_coprime_compact_split_crt.json`. The 520-operation construction above is retained as the fully explicit baseline. Only three source registers are changed:

$$\mathcal{G}_+ = D_2(B)U_c^2 + D_1(B)U_c + D_0(B) + B^{2161}, \qquad L = B^{2279}, \qquad \widehat{N} = \eta\epsilon B^{1201},$$

with the same $\eta = 2^{8\kappa-10}$. In the literal ratio registers use

$$U = \widehat{N}^2 w^2, \qquad Y = \widehat{N}^2 s.$$

Everything else, including the mask, source targets, interval guard, CRT registers, spectator stack, merger, and raw final norm, is unchanged. The JSON register named `N` contains $\widehat{N}$, which can be negative. *Only in the proof*, instantiate the positive Jones parameter by $N = |\widehat{N}|$. Since all uses of this register in the polynomial are through its square, $N^2 = \widehat{N}^2$ exactly. There is no absolute-value gate or extra unknown in the equation.

Unconditional positivity. The same square completion, now with a slightly lower off-target monomial, gives

$$\begin{aligned}\mathcal{G}_+ &> \tfrac{1}{4}B^{1929}U_c^2 + B^{2161} - 4A_*^2B^{2125} - 2A_*B^{1719}\\ &> \tfrac{1}{4}B^{1929}U_c^2 + \tfrac{1}{2}B^{2161},\end{aligned}$$

because $4A_*^2/B^{36} < 1/4$ and $2A_*/B^{442} < 1/4$. Thus

$$S_c > \tfrac{1}{2}N_0B^{1929}U_c^2 + U_c + N_0B^{2161} - \ell > 0$$

for every integer g, exactly as in the baseline proof. Hence $R > L^2$. For the positive proof-side N one has

$$N^2 = 2^{-21}a^{16}bB^{2402} < a^{15}B^{2403} < B^{2418} < B^{4558} = L^2.$$

Also $N \geq c = bB^{116}$, since $\eta B^{1085} \geq 2|\epsilon|$, using $B \geq 2|\epsilon|$ and $\eta \geq 1$. Finally $N \geq B^{1201} \geq 8$. Thus all hypotheses $R \geq N \geq 8$ and $b, c \leq N$ concern the positive number $|\widehat{N}|$, not signed $\widehat{N}$. This is sufficient for every use of the ratio theorem and for the canonical estimate proving $\Delta > 0$.

A smaller block is sufficient. On the required range $0 < g < c$, the previous bound $2|\mathcal{G}| < B^{2162}/256$ still holds. Also $T_* < B^{2162}/256$ and $2B^{2161} < B^{2162}/128$, since $B > 256$. Therefore

$$0 < J_c < B^{2162}/64, \qquad S_c < B^{2279}/32, \qquad T_c < B^{2279}/64.$$

Consequently $S_c + T_c < L = B^{2279}$. The pad is still above every target, since $2161 > 2129$, and the same coefficient bound $|h_e| < 994A_*b^2 < B/8$ covers its extra coefficient 1. The entire sparse carry proof is therefore unchanged.

The exact square threshold uses the forced odd parity. After ratio soundness establishes $b = 2^k$, the polynomial identity $b = 2\epsilon^2$ with integer $\epsilon \neq 0$ forces k odd and $\epsilon^2 = 2^{k-1}$. Write $\nu = \kappa + 2k$. For $L = B^{2279}$,

$$\begin{aligned}\log_2 L + s_2(T_c) + 1 &= 2418\nu - 31k - 21\\ &= 2418\kappa + 4805k - 21\\ &= 2(8\kappa - 10 + 1201\nu) + (k-1) = \log_2 \widehat{N}^2.\end{aligned}$$

Thus $N^2 = \widehat{N}^2$ has exactly the required central-binomial valuation threshold. No even-k assertion is needed: that branch cannot occur. For completeness choose $\epsilon = 2^r > 0$ and $b = 2^{1+2r}$, which are still cofinal. The canonical ω remains a square. The same $b, c \leq N$ estimate proves the interval guard. All later CRT, nonzero, square-recovery, and sign-bootstrap arguments are identical. This proves the compact companion's equivalence to a source zero and hence to a finite ZFC refutation.

Exact degree and accounting. Here the relevant degrees are

$$\deg a_0 = 261194, \quad \deg a_1 = 2634542, \quad \deg C_* = 4874866, \quad \deg \Sigma = 448475,$$

and $\deg A_0 = 5771816$, $\deg A_1 = 3531492$. Again A_0 dominates X^2 and A_1, and the leading term of the output is A_0^2. Its diagonal coefficient is

$$630137\, a^{2885464}\eta^{960} \pmod{1000003} \neq 0.$$

This proves exact degree 11 543 632, with 518 operations under the same gate metric. The change saves two operations and 2480 degrees against the scaled split baseline, without changing the source tuple or number of unknowns. The independent audit checks the all-integer source minimum, compact block bounds, negative-ϵ normalization, odd-k threshold, and that the signed `N` node feeds only its square.

8 A degree-316 ordinary-strong merger

Result and scope. There is an explicit equation in twelve strictly positive integer unknowns, of exact total degree 316, evaluated in 102 binary arithmetic operations (61 multiplications and 41 additions or subtractions), whose positive unsolvability is equivalent in ZFC to Con(ZFC). The three-equation system has residual degrees $2, 26, 288$ in 95 operations, or 94 after the inherited positive transport-coordinate shift. A sum of squares has degree 576 in 103 operations. Fixed numeral leaves and reuse are free; multiplication by a numeral is charged.

This improves the authenticated degree-897, 108-operation twelve-positive construction on both measures. It does not reduce the eleven-positive variable frontier, assert global optimality, prove consistency, or provide a new end-to-end proof-assistant certificate. The native compiler and literal K215/refutation calibration remain explicit inherited dependencies.

The structural change is to retain f and eliminate the ordinary strong coordinate i, together with τ and y. Thus the crucial polynomial is $Q = \Delta(f^2 - 1)$ of degree 14, not the normalized $\Delta^2(ic^2)^2$ of degree 46. A separated three-root sum recovers rationality; rational Pell saturation recovers integrality. Removing a known common denominator factor from its norm gives a degree-288 inner polynomial.

8.1 Literal polynomial

The supplied positive variables are

$$J, F, \alpha, z, f_0, T, s, w, \eta, \zeta, \gamma, t.$$

Use exactly the entire authentic source31 Wang943 121-by-8801 coefficient tuple (β, K_0, M_C, M_F), with $\beta = 2^{1064921} - 1$ and M_F already the shifted mask. These are fixed integers, not unknowns. The payloads, hashes, tuple adapter, and inherited ordinary proof are included in the packet. Every other name below is a polynomial abbreviation:

$$\begin{aligned}
q &= \beta J + 1, & C_w &= q - F - \alpha, & X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & \ell &= EkY,\\
A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac.
\end{aligned}$$

Define

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_t &= (K_0 + w)C_w + q - F - z(q-1),\\
U &= 4q^4, \qquad u = k - R - 1,\\
D_* &= Ub_0 + H(U\gamma - u), \qquad N_m = D_*^2 - U^2 G,\\
f &= f_0 + 1, \qquad Q = \Delta(f^2 - 1), \qquad d = Q - 1,\\
V &= c(Tf - 1) - Rf^2, \qquad E_a = QV^2 - 1, \qquad M_2 = 2f\ell,\\
Z &= c^4 M_2^2 E_a, \qquad x = c^2 t, \qquad r = c^4 f^2 A,\\
M &= d(x^2 + Q + r) - Z,\\
C_0 &= M^2 + 4d^2\{x^2(r - Q) - Qr\}, \qquad B = 2dQ - M,\\
N &= C_0^2 - 16d^2 x^2 r B^2.
\end{aligned}$$

The equation is

$$\boxed{N_t(1 + N_m)(1 + N) - 1 = 0.} \tag{10}$$

The corresponding system is $N_t = 1$, $N_m = 0$, $N = 0$. The sum of squares is $(N_t - 1)^2 + N_m^2 + N^2 = 0$.

The inner norm is exactly

$$N = \prod_{\rho^2=Q,\ \tau^2=A} \left[d(c^2 t - \rho - c^2 f\tau)^2 - c^4 M_2^2 E_a\right], \tag{11}$$

with both formal signs of each root. To verify this identity, put $b = c^2 f\tau$, so $b^2 = r$. Each factor is $M - 2dx\rho - 2dxb + 2d\rho b$. Its norm in ρ is $C_0 + 4dx(2dQ - M)b$; its norm in b is the stated N. This is a polynomial-ring identity, including degenerate specializations. It does not assume independent Galois automorphisms after specialization.

For comparison, the eight-factor linear norm of

$$c^2 dt - d\rho - c^2 df\tau - c^2 M_2 \sqrt{dE_a}$$

is $d^4 N$. Eliminating the two signs of the last root proves this identity and explains the denominator deflation. The actual circuit has no division.

8.2 Unit guard and the main root

Since U is divisible by 4,

$$N_m \equiv D_*^2 \pmod{16}, \qquad N \equiv C_0^2 \pmod{16}.$$

At a root of (10), all three factors are integer units. Neither norm can equal -2, which is not a square modulo 4. Hence $N_m = N = 0$ and $N_t = 1$.

The authentic transport/mask interface then gives

$$q \geq 16, \quad C_w > 0, \quad 3q < R < q^4 - 2, \quad X \geq q, \quad Y \geq q^3, \quad E \geq q^4 > R + 2,$$

$$k \geq 2, \qquad kY < c < k(Y + 1), \qquad a + 2 < 2E.$$

In particular $C_w \leq 0$ would imply $N_t \leq 1 - F - (z-1)(q-1) \leq 0$, so it is impossible. The shift $f = f_0 + 1$ ensures $f \geq 2$, $Q > 1$, and $d > 0$; no $f = 1$ or zero-denominator exception is discarded.

From $N_m = 0$, the rational root $g = D_*/U$ of $g^2 = G$ is an integer. It is positive: $H < 5a$, $b_0 > akY$, and $u < k$ give

$$D_* > Ub_0 - Hk > ak(UY - 5) > 0.$$

Reduction of $D_* = Ug$ modulo U gives

$$U \mid H(k - R - 1). \tag{12}$$

Neither the omitted index quotient nor the old main quotient has yet been recovered.

8.3 Rationality from separated square classes

Write $\rho = \sqrt{Q} > 0$ and $\tau = \sqrt{A} > 0$. If $V = 0$, then $E_a = -1$ and every real factor in (11) is strictly positive. Thus $V \neq 0$. Since V is an integer, $E_a \geq Q - 1 = d$ and

$$y = \sqrt{E_a/d} \geq 1.$$

A vanishing factor gives signs $\epsilon_j \in \{1, -1\}$ with

$$t = \epsilon_1 \frac{\rho}{c^2} + \epsilon_2 f\tau + \epsilon_3 M_2 y. \tag{13}$$

Before any rationality conclusion, one has

$$\rho < (a+2)f, \qquad \tau < \ell + k, \qquad \ell - k = k(EY-1) \geq 2EY - 2 > 2E > a+2.$$

Consequently

$$0 < \frac{\rho}{c^2} < f\ell < f\tau, \qquad \frac{\rho}{c^2} + f\tau < 2f\ell = M_2 \leq M_2 y. \tag{14}$$

Each term in (13) is a rational multiple of a square root of a positive integer; use $y = \sqrt{dE_a}/d$. Distinct squarefree radicals are linearly independent over $\mathbb{Q}$. In a rational signed sum, every irrational square class must cancel. The largest term exceeds the sum of both others, so its class cannot cancel unless y is rational. Then the strict inequality between the two remaining terms excludes their cancellation in an irrational class. Hence ρ, τ, y are rational. In particular ρ and τ are integers. This argument explicitly allows coincident square classes.

Rational Pell saturation. For integers $m \geq 2, V$ and rational y, the equation

$$m^2V^2 - (m^2-1)y^2 = 1$$

forces y to be integral. Indeed, write $y = b/r$ in lowest terms. Then $r^2 \mid m^2-1$. For the positive nonsquare integer $D_0 = (m^2-1)/r^2$, the pairs (m, r) and $(m|V|, |b|)$ are integer-coordinate Pell solutions. If their fundamental-unit indices are j, l, divisibility of their abscissas forces l to be an odd multiple of j: reduce modulo $2j$, reflect into $[0, j]$, and use strict growth with $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$. Ordinate divisibility then gives $r \mid b$, so $r = 1$. The original equation excludes zero-coordinate cases.

Apply this lemma with $m = \rho$ to

$$\rho^2V^2 - (\rho^2-1)y^2 = 1.$$

Thus y is integral. Equation (13) now makes ρ/c^2 integral. Put $i = \rho/c^2 > 0$. The recovered equations are

$$\tau^2 = 1 + \ell(\ell+k), \qquad (ic^2)^2 = \Delta(f^2-1), \qquad QV^2 - (Q-1)y^2 = 1.$$

All square-root signs may be made positive. No cancellation of Δ and no ordinary-rank assumption has occurred in this recovery.

8.4 Ordered index recovery

This part uses the preserved ordinary strong-rank theorem and the source50 absorber, with their hypotheses now verified. Let $A_0 = a+2$ and $P_0 = 2EY+1$. The two recovered Pell norms yield indices $p, n \geq 1$ with

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \quad \tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n).$$

The first norm's fundamental integer-coordinate solution is $(2EY+1, 2)$ for discriminant $EY(EY+1)$. Since $P_0 > A_0$ and $c > k$, monotonicity gives $p > n$.

The Pell recurrence gives $k \equiv 2n \pmod{4EY}$, and $4EY = 4ws^2q^7$ is divisible by U. Set $\mu = U/\gcd(U, H)$. The number H is odd; a prime common to q, H must be 3. If $3 \mid q$, then $H \equiv 3 \pmod{27}$, so its 3-adic valuation is exactly one. Hence

$$\gcd(U, H) \leq 3, \qquad \mu \geq \frac{4q^4}{3} > R+1.$$

Together with (12) this gives $2n = R + 1 + j\mu$, where positivity of n forces $j \geq 0$. Thus $n \geq 25$, $p \geq 26$, and Pell growth proves

$$c > A_0\Delta^2, \qquad c > 2p, \qquad c > 2R.$$

The ordinary strong-rank theorem now applies to $(ic^2)^2 = \Delta(f^2-1)$. It gives an index m such that

$$f = \chi_{A_0}(m), \quad p \mid m, \quad c \mid m, \quad m > 2p, \quad ic^2 = \Delta\psi_{A_0}(m), \quad c^2 \mid f^2 - 1.$$

The theorem handles the possibly nonfundamental parameter in the underlying squarefree Pell field. Its proof and the signed auxiliary argument are included in the preserved ordinary source, Sections 4–5.

The odd auxiliary index j_0 satisfies $j_0 \equiv \pm p \pmod m$ from $V \equiv -c \pmod f$, and $j_0 \equiv \pm R \pmod c$ from $Q \equiv 0 \pmod c$, $V \equiv -R \pmod c$. Since $c \mid m$ and $0 < p, R < c/2$, it follows that $p = R$. The signed reduction uses $2\chi_{A_0}(m-1) < \chi_{A_0}(m)$, not an assumed sign of V.

Now $n < R$, so $|2n - R - 1| < R < \mu$ forces $2n = R + 1$. Hence $u = k - 2n$ is divisible by $4EY$, and therefore by both E, U. Set

$$h = u/E, \qquad \gamma_{\mathrm{old}} = \gamma - u/U.$$

The first is positive because $n \geq 25$ and $\psi_{P_0}(n) > n$. The main carrier gives $\gamma_{\mathrm{old}} = (g - b_0)/H$. Since $n \geq 2$, $k \geq 4P_0 > X$, so $c > X$ and $g - b_0 > (a+1)c - ac - X = c - X > 0$. All positive ordinary compiler coordinates are restored. Its inherited soundness theorem extracts a blank K215 halt and a finite ZFC refutation.

8.5 Completeness

Conversely take a canonical positive ordinary compiler witness. Its index identity is $p = R = 2n - 1$, so u is a positive multiple of $4EY$ and of U. Set

$$\gamma = \gamma_{\mathrm{old}} + u/U, \qquad f_0 = f - 1, \qquad t = i + f\tau + M_2 y.$$

These are positive integers. Then $D_* = Ug$, while the positive-sign factor in (11) is $c^4 M_2^2(dy^2 - E_a) = 0$. Thus $N_m = N = 0$ and $N_t = 1$. The preserved change $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$ provides ordinary witnesses from normalized ones when necessary. This proves existence equivalence, hence the claimed consistency equivalence.

The shifted system uses $(K_0 + w)C_w = F + z(q-1)$, corresponding to $z_{\mathrm{old}} = z + 1$. Canonical witnesses have $X = 2^R$ and $w > q > F$, which force $z_{\mathrm{old}} \geq 2$. Thus the new z is positive, and the shift saves one operation.

8.6 Exact degree and replay

Put

$$q_0 = \beta J, \quad Y_0 = sq_0^3, \quad E_0 = wq_0Y_0, \quad c_0 = (\eta + \zeta)Y_0.$$

These have degrees $1, 4, 6, 5$. The inner leading forms are

$$Q_{14} = E_0^2 f_0^2, \quad V_7 = c_0 T f_0, \quad (M_2)_{12} = 2f_0 E_0 c_0, \quad Z_{72} = 4E_0^4 c_0^8 T^2 f_0^6.$$

All other terms of M have degree at most 58. Therefore

$$N_{288} = Z_{72}^4 = 2^8 E_0^{16} c_0^{32} T^8 f_0^{24}.$$

The subtracted term in N has degree at most 238, so there is no possible cancellation of this highest form. For the main residual use

$$N_m = U^2\{(b_0 + H\gamma)^2 - G\} - 2UHu(b_0 + H\gamma) + H^2u^2.$$

The bracket has degree 18 and highest form $8E_0^2 c_0\gamma$; the other terms have degree at most $25, 20$. Thus

$$(N_m)_{26} = 128 q_0^8 E_0^2 c_0 \gamma.$$

Since $(N_t)_2 = w(q_0 - F - \alpha) - zq_0$ is nonzero,

$$P_{316} = 2^{15} q_0^8 E_0^{18} c_0^{33} \gamma T^8 f_0^{24} [w(q_0 - F - \alpha) - zq_0]$$

is nonzero for every authentic $\beta > 0$. These are exact fixed-instance degrees, not generic numerical tests.

The packet's independent arithmetic checker verifies 36 exact signed norm/deflation identities, 100 modular identities, both residue guards, 144 pre-index bounds, and 24 complete finite-field polynomial rays including the actual fixed coefficients. It also verifies payload hashes, analytic leaders, and transport shifts. Both ordinary Python and `python -O` runs pass. The inert circuits have no dead gates or inputs. Finite checks validate identities, counts and degrees, not the inherited unbounded compiler theorem. The work is AI-assisted and makes no claim of independent human review or kernel-certified consistency reduction.

9 A synthesized short-polynomial frontier

The squared-modulus quotient, ordinary strong-rank theorem, and two Pell-root gaps are all inherited. Their combination improves several local degree/operation tradeoffs, without reducing the absolute operation record below the incoming 66-operation, degree-157 equation. All equations below have strictly positive integer unknowns, and their lack of solutions is provably equivalent in ZFC to Con(ZFC) at the ordinary mathematical proof level of the inherited compiler. This is not a new end-to-end Lean/Rocq certificate or a consistency assertion.

Variables	Operations	Exact degree	Construction
15	67	125	Reduced scaled product; squared modulus
15	68	101	Ordinary product; squared modulus
15	69	91	Ordinary product; first gap
15	70	84	Ordinary product; both gaps
15	71	80	Three squared residuals
15	72	74	Three squared residuals; first gap
15	73	60	Four squared residuals
15	74	50	Four squared residuals; auxiliary gap
15	75	46	Four squared residuals; both gaps
14	79	52	Absorber; squared modulus; auxiliary gap

Comparison discipline. The incoming 15-variable 66/157 and 76/44 points remain incomparable endpoints. Historical pairs 67/155, 68/123, 69/99, 70/93, 71/83, 74/56 and 78/44, together with incoming 68/113 and wave-7 72/80, are retained as comparators. The 14-variable point improves the incoming 79/56 construction at the same operation count. The included 76/44 reconstruction is a reference control, not new work. These are bounded repository comparisons, not global minimality or priority claims.

9.1 Literal fixed polynomials

Use the entire Wang943 121×8801 tuple, with $B = 2^{1064921}$, $\beta = B - 1$, $K_0 = K$, $M_C = M_C$, and $M_F = M_{F,\text{source}} = M_{F,\text{native}} + B - 1$. All four materialized coefficients and their compressed/raw hashes accompany the report. In particular, M_F is already shifted. For fifteen coordinates use

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma > 0.$$

Replace τ by r when the first gap is enabled and y by u when the auxiliary gap is enabled. Define polynomial abbreviations

$$q = \beta J + 1, \qquad C = q - F - \alpha, \qquad X = wq, \qquad Y = sq^3,$$
$$E = XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta, \qquad a = E + Y,$$
$$H = 4a + 3, \qquad \Delta = a^2 + H, \qquad D = X + ac + \gamma H, \qquad L = EkY,$$
$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,$$
$$V = f^2(cT - R) - c, \qquad A_0 = \Delta(f^2 - 1), \qquad I = ic^2.$$

They introduce no additional unknowns. The norm factors are

$$N_0 = \tau^2 - L(L + k), \qquad N_1 = D^2 - \Delta c^2,$$
$$N_a = A_0(V^2 - y^2) + y^2, \qquad N_k = k - hE - R,$$
$$N_t = (K_0 + w)C + (q - F) - z(q - 1), \qquad S = I^2 - A_0, \quad N_s = S + 1.$$

The gap alternatives are

$$N_{0,g} = r^2 + L(2r - k), \qquad N_{a,g} = (V + u)^2 - A_0 u(2V + u).$$

Let $\mathcal{Q}(B_1, \ldots, B_m) = S^2 + \sum_j (\prod_{N \in B_j} N - 1)^2$. The displayed data specify the ordinary polynomials completely:

Operations	Equation's left side	Enabled gaps
68	$N_0 N_1 N_a N_k N_t N_s - 1$	None
69	$N_0 N_1 N_a N_k N_t N_s - 1$	First
70	$N_0 N_1 N_a N_k N_t N_s - 1$	Both
71	$\mathcal{Q}(\{N_0, N_1\}, \{N_a, N_k, N_t\})$	None
72	$\mathcal{Q}(\{N_0, N_1, N_k\}, \{N_a, N_t\})$	First
73	$\mathcal{Q}(\{N_0\}, \{N_a\}, \{N_1, N_k, N_t\})$	None
74	$\mathcal{Q}(\{N_a\}, \{N_1, N_k\}, \{N_0, N_t\})$	Auxiliary
75	$\mathcal{Q}(\{N_a\}, \{N_0, N_k\}, \{N_1, N_t\})$	Both

For the 67-operation equation only, put $I_{\mathrm{sc}} = i\Delta c^2$ and $N_{\mathrm{sc}} = \Delta f^2 - I_{\mathrm{sc}}^2$. With no gaps, its left side is

$$N_0 N_1 N_a N_k N_t N_{\mathrm{sc}} - \Delta.$$

The evaluator shares Δf^2 between N_{sc} and $A_0 = \Delta f^2 - \Delta$. Every binary operation, including fixed-scalar multiplication, is charged in the literal JSON gate lists.

9.2 Why the combinations preserve existence

The exact all-value pullback is

$$c((fT)f - 1) - Rf^2 = f^2(cT - R) - c.$$

Thus a new positive quotient gives the old positive quotient $T_{\mathrm{old}} = fT$. Completeness is not the assertion that every old quotient is divisible by f. Instead, at a canonical compiler witness, write $A = a + 2$ and $c = \psi_A(R)$, where $R \equiv 3 \pmod 4$. Ordinary strong rank supplies

$$I^2 = \Delta(f^2 - 1), \qquad c \mid I, \qquad f^2 \equiv 1 \pmod c.$$

For $R = 2j + 1$ the integer polynomial Q_j defined by $\chi_t(R) = tQ_j(t^2)$ satisfies

$$Q_j(0) = (-1)^j R, \qquad Q_j(1 - A^2) = (-1)^j \psi_A(R).$$

The recurrence $Q_{j+1} = (4Z-2)Q_j - Q_{j-1}$, with $Q_0 = 1$ and $Q_1 = 4Z-3$, proves both identities. Since j is odd, set

$$V_* = \frac{\chi_I(R)}{I}, \qquad y_* = \psi_I(R).$$

The quotient is integral, $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod{c}$. Since $\gcd(c, f) = 1$, the number

$$T_* = \frac{V_* + c + Rf^2}{cf^2}$$

is a positive integer and satisfies the new quotient formula. The auxiliary norm remains one. This argument works with the ordinary strong coordinate; it never divides an arbitrary supplied i by Δ.

The root gaps use exact polynomial identities:

$$(L+r)^2 - L(L+k) = r^2 + L(2r-k),$$
$$A_0\big(V^2 - (V+u)^2\big) + (V+u)^2 = (V+u)^2 - A_0u(2V+u).$$

For soundness restore $\tau = L + r > 0$ and $y = |V+u|$. Once $A_0 = I^2 > 1$, an auxiliary unit excludes $V + u = 0$. For completeness $r = \tau - L > 0$ follows from $\tau^2 - L^2 = Lk + 1 > 0$. The canonical $V_* > 1$ and

$$(I^2-1)(y_*^2 - V_*^2) = V_*^2 - 1 > 0$$

give $u = y_* - V_* > 0$.

For the ordinary product, all factors are units. The full mod-four check shows $N_s = 1 + I^2 - \Delta(f^2-1) \not\equiv -1 \pmod 4$, since $\Delta \equiv 0$ or 3 and squares are 0 or 1. Hence $S = 0$; in the squared sums this is immediate. The consecutive-product negative Pell obstruction excludes $N_0 = -1$; modulo four excludes $N_1 = -1$ and, now that $A_0 = I^2$, $N_a = -1$. Thus

$$N_0 = N_1 = N_a = 1, \qquad N_kN_t = 1, \qquad S = 0.$$

Every grouping accounts for each of the five factors once, so these values pull back to the inherited relaxed 72-operation polynomial. Its proof retains the possible $C = 0$ boundary, establishes the large main index before invoking ordinary rank, recovers $p = R$, and only then forces $N_k = N_t = 1$ and $C > 0$. The authenticated native compiler and blank-K215 calibration supply the finite ZFC refutation. Conversely, a canonical compiler witness makes all individual factors one after the replacements.

For the scaled equation cancel positive Δ and apply the normalized strong-unit mod-four obstruction. Then $N_{\mathrm{sc}} = \Delta$ and $A_0 = I_{\mathrm{sc}}^2$, reducing to the inherited scaled proof. The canonical completeness argument above applies with I_{sc} in place of I. Full local arguments and inherited interfaces are included in `PROOF.md` and the authenticated source-proof directory.

9.3 The fourteen-variable degree reduction

Remove h, retain τ, and replace y by u. Put

$$U = 4q^4, \quad r_0 = k - R - 1, \quad b_0 = X + ac, \quad D_* = Ub_0 + H(U\gamma - r_0), \quad G_* = 1 + \Delta c^2.$$

The equation is

$$(D_*^2 - U^2G_*)^2 + S^2 + (N_{a,g} - 1)^2 + (N_0N_t - 1)^2 = 0.$$

It is an exact existence adapter of the incoming 14-variable 79/56 absorber: $T_{\mathrm{old}} = fT$ and $y = |V+u|$ give soundness directly. The absorber proof's early index bound and ordinary rank theorem are inherited, not replaced. Canonical squared-modulus reconstruction followed by $u = y_* - V_*$ supplies completeness. One saved quotient gate pays for one added auxiliary-gap gate, leaving 79 operations.

The exact main residual identity is

$$D_*^2 - U^2 G_* = U^2((b_0 + H\gamma)^2 - G_*) \\ - 2UHr_0(b_0 + H\gamma) + H^2 r_0^2.$$

Its three terms have degrees $26, \le 25, \le 20$; the other residuals have degrees $22, 23, 24$. Therefore the exact final degree is 52, not 44. With $Q = \beta J$ and $k_0 = \eta + \zeta$, its nonzero leading form is $16384\gamma^2 w^4 s^6 k_0^2 Q^{38}$.

9.4 Arithmetic verification and proof boundary

The independent standard-library auditor imports no generator or upstream Python code. It expands each literal main-norm ancestor cone exactly, then propagates symbolic leading forms, preserving cancellations rather than using only naive degree bounds. The factor degrees are

$$(\deg N_0, \deg N_1, \deg N_a, \deg N_k, \deg N_t, \deg S) = (22 \text{ or } 12, 18, 30 \text{ or } 23, 7, 2, 22).$$

The scaled factor has degree 46. All output leading forms have an exhibited monomial with nonzero coefficient depending only on $\beta > 0$, so the proof applies to the actual bound tuple. All gates, unknowns and ports are live. Raw/compressed coefficient hashes, ranges, mask populations and source-proof hashes are checked, along with five universal local polynomial identities. Each of the eleven circuits passes 100 signed/degenerate literal-formula diagnostics; five exact canonical Pell examples check reconstruction and positive gaps. Checks pass normally and under `python3 -O`.

These are finite artifact checks, separate from the unbounded Pell and compiler proofs. No finite experiment tests Con(ZFC). The contribution is the explicit combined frontier, reproducible circuits and reduction proof, not a claim of a foundational breakthrough.

10 Proof status and relationship to the formal developments

At the reviewed placement snapshot, the repository reports a closed executable `ExplicitZFC` expression, a direct arithmetic-graph `NativeZFC` equivalence, and a `PositiveZFC` natural-to-positive coordinate shift. Numerical coefficient payloads of those formal expressions remain unmaterialized. The generic quadratic extraction foundations are reported checked, but their complete adaptation to the final Jones theorem is separate. The object-language ZFC development has actual addition formulas and derivations, while its full internal consistency bridge is still open. We inspected this status; we did not rerun those Lean builds.

Those results are not kernel certificates for the smaller fixed circuits in this report. Conversely, the existence of noncomputable historical polynomial selections does not make all newer executable constructions noncomputable. We distinguish explicit coefficient data, external Lean semantic equivalences, object-language derivations, and ordinary proofs formalizable in ZFC. None may be substituted for another without an actual bridge theorem.

The literal K215 logical calibration, native compiler, Pell rank and Jones ratio results are identified inherited dependencies. Only the ten-integer route additionally uses Sun's 2026 bridge. Local finite tests certify exact identities, bindings, circuit sizes and degree attainment; they do not certify an unbounded compiler by testing a finite number of assignments.

11 Validation and reproducible archive

The root verifier authenticates the complete release, copies it into a temporary directory, and executes 25 finite replay stages without modifying the sealed originals. The stages cover kernel reconstruction, source and circuit identities, independent literal evaluators, exact degree witnesses, coefficient bindings,

guard counterexamples and a cross-family metadata check. Generated JSON circuits are compared byte-for-byte with the delivered versions. The machine-readable stage and output lists are authoritative.

The standard root run uses ordinary Python. The nine-variable primary verifier deliberately rejects `-O` and `-OO`; that rejection is documented and was tested, not misreported as a successful optimized proof. Its separate independent checker uses explicit checks and passes in both modes. Other component receipts distinguish normal and optimized runs. NumPy is required for one direct-merger diagnostic; the other canonical families and the direct merger's independent symbolic checker use the Python standard library. The late thirteen-positive authored checker additionally uses SymPy; its separate independent review is standard-library only. Tested versions are Python 3.12.14, NumPy 2.3.5 and SymPy 1.14.0.

The strongest local checks are exact identities, not sample evaluations. The split-source checker parses the original 22 residuals, applies and inverts the two row operations, authenticates all 89 coefficient records, and verifies all 22,544 interactions and 22 formal target identities. The direct merger's independent checker expands its complete formal norm, including zero, negative and coincident specializations. Degree certificates combine valid upper bounds with nonzero leading homogeneous forms or specializations modulo primes; merely evaluating random points would not establish exact degree.

Finite examples additionally stress source bounds, signed borrows, zero spectators, CRT parametrizations, and positive-coordinate reconstruction. Ten deliberately corrupted source/code cases are rejected. Three separate root corruption controls test release integrity only. These distinctions are retained in the receipts: a file-hash rejection is not an independent number-theoretic proof, and a finite test is not a consistency decision.

All primary source31 bindings agree on the same four actual payloads. The nine-variable comparison tuple and the late thirteen-positive compact tuple are explicitly separate. Active proof and payload paths are checked after copying to an isolated directory; historical metadata is identified as inert provenance. No network access is needed for replay. The imported machine/compiler and representation results retain immutable references and original notices; the replay does not claim to rebuild their theorem-prover developments.

The PDF is compiled from the supplied TeX, inspected page by page, and packaged with the explicit circuits, coefficient payloads, proofs and checks. Rebuilding a PDF may change its metadata. The final archive is re-extracted and replayed before delivery; its checksum is supplied outside the archive to avoid a self-referential checksum.

12 Limits, rejected shortcuts, and further targets

The new code is certified, not proved optimal. Its search history is not a semantic dependency: the finite tuple and exact isolation checker are sufficient. More aggressive fixed-port row operations or a different source presentation could improve it further, but require new coefficient, positivity and degree audits.

No twelve-unrestricted-integer theorem or ten-positive-variable theorem is claimed. In the signed construction the zero-spectator branch is handled by recovering the first two guards and using source positivity for every signed packed input before the final guard is read. Omitting that ordering would permit an invalid argument. In the ordinary-root merger, integrality comes from Pell saturation followed by the carrier identity. A tempting shortcut from $c^2 \mid d\rho$ and $\gcd(d, \rho) = 1$ to $c^2 \mid \rho$ is invalid and is not used.

The most useful formalization targets are the degree-specific extraction certificate, sparse-borrow normalization, and the ordinary-root merger's ordered saturation proof. They are reusable arithmetic interfaces rather than assertions about consistency itself. The supplied source maps, explicit gates and independent algebraic checkers are intended to make such work easier, not to present it as already completed.

13 Late parallel releases and the moving frontier

After the eighteen-stage replay had passed, two further original archives became available during final packaging. They are not part of the earlier 46–53 snapshot. Their delivered metrics change the interpretation of the comparison above; we do not claim an exclusive discovery or a global record.

Construction	Degree	Operations
Separated Degrees: 9 positive	2 628 944	499
This packet, initial: 9 positive	2 676 548	476
Separated Degrees: 10$\mathbb{Z}$	8 317 400	544
This packet: 10$\mathbb{Z}$	8 469 528	533
Separated Degrees: 12 positive	242	94
Separated Degrees: 12 positive	258	89
This packet: 12 positive	316	102
Free Coordinate: 15 positive	54	73
Free Coordinate: 15 positive	46	74

The initial nine- and ten-variable points therefore become degree/cost tradeoffs: our evaluators save 23 and 11 operations, respectively, but the parallel polynomials have slightly smaller degrees. The parallel twelve-positive points dominate our moderate-degree equation on both metrics. We retain our construction as a distinct, completely specified ordinary-root merger, not as the current best twelve-positive equation. The parallel short points also supersede some of our short frontier entries.

The parallel degree-separated source uses 94 records with maximum weight108 and a different whole native coefficient tuple. Our independently developed row shear gives 89 records and maximum weight116. These are not interchangeable certificates or individual coefficient ports. Their shared conceptual idea is to stop separating irrelevant cross-degree collisions. The parallel free-coordinate proof has now been composed with our source in the separately pinned supplement below. Its independently audited nine-positive equation has degree 1 910 404 and 473 operations, improving both metrics over the late nine-variable comparator. The natural translation costs480. This is a new literal circuit, not a retrospective change to the initial 476-operation evaluator. A second supplement supplies a thirteen-positive degree126/91 quotient tradeoff against the late degree132/83 equation.

The original late archives, hashes, scoped reviews and replay outcomes are identified in `review/`. Their delivered claims are distinguished from our own kernel-free mathematical checks. No new proof-assistant endpoint is inferred from either set of finite checks.

The repository catalogue was subsequently reconciled at revision `a71325bf18a4`; its current review and catalogue are copied alongside the earlier status snapshot. That later review also highlights the need to justify cofinal square witnesses from the actual Jones construction, rather than its bare existential theorem statement. Our signed proof cites the chosen witness and its preservation through the ratio adapter explicitly.

14 Free-coordinate composition: nine variables of degree 1,910,404

A late supplied FreeCoordinate contribution combines with the preceding split116 source to improve both metrics: the new equation has nine strictly positive unknowns, exact total degree 1 910 404, and 473 live binary operations (162 additions, 22 subtractions, 289 multiplications). The nonnegative-domain translation has the same degree and 480 operations. This is a separate polynomial, not an edit to the frozen preceding family. The supplied contribution is credited for the free carrier, quotient shift, and deletion of the leading block; the composition and its exact circuits are newly checked here. Its proof and review are retained verbatim and pinned at source snapshot `4b4349064a750ba6347293b55874a7f92abdd0db`. The same explicit compiler and Jones dependencies, consistency interpretation, and limitations stated above apply.

14.1 The smaller outer rank and its exact calibration

Keep the complete original31 tuple, all split116 kernels, and $a = 2^{2134939} = 4096M$, $M = 2^{2134927}$. Keep the preceding definitions of $B = ab^2, \ell, U_s, G_s, A_q, N_0, J_c, S_c, T_c$, and $G_{\mathrm{cap}} = 32bB^{116}$. Replace only the outer registers by

$$L = bB^{2279}, \qquad R = L(S_c + T_c + 1) - S_c - 1, \qquad c = 2^{17079501}, \quad N = cbB^{1201}.$$

The previous unconditional source positivity remains valid for every $b, g \geq 1$. Therefore

$$R = (L-1)S_c + LT_c + L - 1 \geq L - 1 \geq 8, \qquad b \leq R, N, \quad N \geq 8.$$

On the full soundness branch $g < 4bB^{116}$, not merely a two-word bound, $U_s < 6bB^{116}$ and

$$\frac{2G_s}{B^{2162}} < \frac{86Mb^2}{B-1} < \frac{1}{32}, \qquad A_q, (B-2)A_q < \frac{B^{2162}}{64}.$$

Consequently $S_c + T_c < B^{2279} \leq L$, strictly even when $b = 1$. Deleting the old leading L^2 thus leaves two disjoint binary blocks:

$$R = (S_c + T_c)L + (L - 1 - S_c).$$

For $b = 2^k$, $B = 2^v$, $v = 2134939 + 2k$, put $u = k + 2279v$. The carry defect $\tau = \mathrm{pop}(S_c) + \mathrm{pop}(T_c) - \mathrm{pop}(S_c + T_c)$ is nonnegative and vanishes exactly when $S_c \,\&\, T_c = 0$. Hence

$$\begin{aligned} \mathrm{pop}(R) &= u + \mathrm{pop}(T_c) - \tau, \\ u + \mathrm{pop}(T_c) &= 2418v - 30k - 22 = 2(17079501 + k + 1201v) = 2\log_2 N. \end{aligned}$$

The exact identity uses $16 \cdot 2134939 - 22 = 2 \cdot 17079501$. Legendre's formula again gives $N^2 \mid \binom{2R}{R}$ exactly when the source masks are disjoint. The case $k = 0$ is included in the calibration; it cannot give a zero, because then the low mask is $N_0 - 1$ and $0 < g < N_0$. The unchanged sparse-digit proof recovers precisely the positive common source zeros. Every source zero admits sufficiently large binary b and gives $32g < G_{\mathrm{cap}}$.

14.2 A free carrier and a positive shifted numerator

The nine positive inputs are now

$$b, g, h, s, w, C, i, j, n \geq 1, \qquad j_0 = j - 1 \geq 0.$$

Define the ratio registers by

$$\begin{aligned} U &= N^2 w, & Y &= N^2 s, & M_0 &= RY, \\ P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1), \\ B_1 &= 2R + 1, & W_1 &= bw, & J &= 4A - 5, \\ Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2-1)C^2 + 1, & k_0 &= iJC, \\ E &= k_0 DC, & F &= (A^2-1)E^2 + 1, & G &= A + F(CD + 1 - A), \\ H &= C + F(B_1 + j_0 C), & I &= (G^2-1)H^2 + 1, & & \\ a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI. & & \end{aligned}$$

The normalized margin is unchanged:

$$t = bB^{116}, \qquad m = K^2(4t - g) - 16t(C - KY)^2.$$

Here $8m$ is the original Jones margin. Although free C no longer makes Q positive, the shifted numerator is unconditionally positive:

$$E_0 = J(n + W_1^2) + Q = Jn + (J-2)W_1^2 + 3W_1C + 2 > 0,$$

because $J \geq 3$. Once $m > 0$, the carrier range is recovered:

$$|C - KY| < K/2, \qquad C > K(Y - 1/2) > 2K \geq 2(P + R) > 2R + 1 + W_1.$$

We used $Y \geq 64$ and $P > U \geq W_1$. Thus the old coordinate $\phi = C - B_1 - W_1$ is positive. Exact CRT transport, with $F \equiv 1 \pmod{J}$ and $F \equiv 1 \pmod{C}$, gives

$$j_{\text{old}} = 1 + \frac{F-1}{C} B_1 + F j_0 \geq 1.$$

Conversely, the positive quotient $(H_{\text{old}} - C)/F$, congruent to $B_1 \pmod{C}$, is uniquely $B_1 + j_0 C$ with $j_0 \geq 0$, since $0 < B_1 < C$. Retaining $j_0 = j - 1$ is essential; no cofinality assertion about positive unshifted j is used.

14.3 The equation and both directions

The same unconditional strict majorants work for every free $C \geq 1$:

$$\begin{aligned} V_0 &= PK, & T_* &= k_0(8D^2 - 4D - 1), \quad V_1 = T_* GH, \\ V_0^2 - a_0 &= K^2 - 1 > 0, \\ T_*^2 - 64DF &= (8D+1)k_0^2 - 64D > 0, \\ \Delta_1 := V_1^2 - 64a_1 &= \big((8D+1)k_0^2 - 64D\big)(GH)^2 + 64DF(H^2 - 1) > 0. \end{aligned}$$

Put $\Omega = Jm$, $\Theta = E_0 - \Omega(V_0 + V_1)$. The new equation is

$$\boxed{F_9 = \big[\Theta^2 - \Omega^2(a_0 + 64a_1)\big]^2 - 256\Omega^4 a_0 a_1 = 0.}$$

If $m = 0$, its value is $E_0^4 > 0$. Otherwise the real four-factor norm identity gives signed roots $r_0^2 = a_0, r_1^2 = 64a_1$ with

$$n + W_1^2 + Q/J = m(V_0 + V_1 + r_0 + r_1).$$

The left side and the parenthesis are strictly positive, so $m > 0$ follows before root integrality. Recover the carrier range above; then $a_1 > C^2A^4 > K^2P^2 > a_0$. Thus the rational sum $x = r_0 + r_1 = \Theta/\Omega$ is nonzero, and $2xr_0 = x^2 + a_0 - 64a_1$ proves both roots rational, hence integral. Also $r_1/8$ is integral. The factor identity now gives $J \mid Q$. The Jones interface therefore returns binary b, $8g < G_{\text{cap}}$, and the binomial divisibility, proving soundness.

For completeness use the source and ratio witnesses and positive roots. The bounds $A > W_1$, $D \geq A^2$, $D > C^2$ imply $W_1^2 < D$ and $W_1 C < D$. Since $Q/J \leq W_1 C$ and $T_* \geq 9D$,

$$W_1^2 + Q/J < 2D < V_1.$$

The normalized integer margin satisfies $m \geq 1$, so

$$n = m(V_0 + V_1 + r_0 + r_1) - W_1^2 - Q/J > 0$$

is an admissible integer and annihilates one norm factor. This proves the claimed positive-domain equivalence, including the shifted final witness.

14.4 Exact degree and reproducible certificates

The source heads are $\deg S_c = 4556$, with leading term $2a^{2278}$ on the diagonal, and $\deg T_c = 4494$. Hence $r = \deg R = 9115$, with head $2a^{4557}$, while $q = \deg N = 2403$, with head ca^{1201}. The separate certificate resolves the inner cancellation using

$$\Theta^2 - \Omega^2(a_0 + 64a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\big[(K^2 - 1) + \Delta_1 + 2V_0V_1\big].$$

Its uniquely dominant term is $\Omega^2\Delta_1$, of degree 955202; the final subtracted norm term has degree 1385948. Therefore

$$\deg F_9 = 104r + 400q + 8 \cdot 116 + 316 = 1\,910\,404, \qquad \mathrm{lc}_{\mathrm{diag}}(F_9) = 2^{186}a^{954792}c^{400} > 0.$$

The diagonal attains the multivariate upper bound for the actual tuple. Replacing all positive inputs by natural inputs plus one preserves this degree and the zero-set equivalence; internally j_0 is then the natural input j.

The adjunct supplies four exact DAGs, two degree certificates, the full coefficient tuple, provenance pins, a standard-library builder/checker, and two separate independent AI-assisted reviews. The checks authenticate 36 imported files, verify 22544 extraction interactions, exactly expand the source kernels, replay actual-tuple residues, and confirm all changed register identities, the 473-operation ledger, and the nonzero degree head. The binary and finalizer fixtures are finite implementation checks, not substitutes for the unbounded proof or imported numerical theorems.

15 A thirteen-positive degree-126 auxiliary quotient

This additional construction has thirteen positive unknowns, exact degree 126, and 91 binary arithmetic operations (51 multiplications and 40 additions/subtractions). It is a degree–operation tradeoff against the late incoming degree-132, 83-operation construction, not a dominance claim. It uses that construction's entire fixed native-window coefficient tuple, with $\beta = 2^{590069} - 1$ and the already shifted M_F.

Supply exactly $J, F, \alpha, z, f, i, T, s, w, \eta, \zeta, y, t > 0$. Define

$$\begin{aligned}
&q = \beta J + 1, && X = wq, && Y = sq^3, && E = XY,\\
&k = \eta + \zeta, && c = kY + \eta, && a = E + Y, && H = 4a + 3,\\
&\Delta = a^2 + H = (a+1)(a+3), && \ell = EkY,\\
&A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2, && b_0 = X + ac.
\end{aligned}$$

Set

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_t &= (K_0 + w)(q - F - \alpha) + q - F - z(q-1),\\
U &= 4q^4, \qquad r = k - R - 1, \qquad S = UH,\\
W &= Ub_0 + H\big(U(t + \ell) + r\big),\\
N &= (W^2 + U^2G - S^2A)^2 - 4W^2U^2G.
\end{aligned}$$

These abbreviations are the late parent's separate main/first-Pell norm. The new auxiliary abbreviations are

$$\begin{aligned}
&I = ic^2, && N_s = f^2 - \Delta I^2, && Q = \Delta(f^2 - 1),\\
&V = c(Tf - 1) - Rf^2, && M = a + 1, && C_* = (a+3)(f^2 - 1),\\
&Z = C_*(V^2 - 1) + (Q - 1)y(2 - My).
\end{aligned}$$

In particular $MC_* = Q$; the notation C_* is unrelated to the fixed coefficient port M_C. The single equation is

$$\boxed{P = N_t(1+N)N_s(1+Z) - 1 = 0.}$$

Every abbreviation is polynomial and introduces no quantified unknown.

The quotient identity and the sign gap. Direct expansion gives the universal polynomial identity

$$MZ = QV^2 - (Q-1)(My-1)^2 - 1. \tag{*}$$

For integers $m \geq 2$ and $0 < b < 2m-2$, the equation $x^2-(m^2-1)v^2 = -b$ has no integer solution. Indeed, after taking absolute values, $v > 0$ and $(m-1)v < x < mv$, because $(2m-2)v^2 > b$. Multiplication by the norm-one element $m-\sqrt{m^2-1}$ gives another solution with strictly smaller positive ordinate $mv-x$. Infinite descent proves the assertion.

Soundness, with the order explicit. If $P = 0$, its four integer factors are units. Since N is a square modulo 4, the factor $1+N$ must be 1. The negative choice for N_s is excluded by the preceding lemma with $m = a+2$ and $b = 1$, since $\Delta = (a+2)^2 - 1$. Thus $N_s = 1$ and $Q = (\Delta I)^2$. Put $m = \Delta I \geq 2$. If $1+Z = -1$, identity (*) gives

$$(mV)^2 - (m^2-1)(My-1)^2 = -(2a+1).$$

This is impossible because $0 < 2a+1 < 2\Delta I - 2$: already $I \geq 1$, $a > 0$, and $2\Delta - 2 - (2a+1) = 2a^2 + 6a + 3 > 0$. Therefore $Z = 0$ and $N_t = 1$. Notice that no transport bound was needed before this recovery. Now $y_{\text{old}} = My - 1 > 0$ restores exactly the late parent's thirteen-positive system:

$$N_t = 1, \quad N = 0, \quad N_s = 1, \quad QV^2 - (Q-1)y_{\text{old}}^2 = 1.$$

The parent's main-norm recovery and normalized strong/auxiliary rank proof therefore restore the full compiler interface, giving a blank K215 halt and a finite syntactic ZFC refutation.

Completeness and the residue. Start with canonical normalized compiler witnesses for an accepting computation, keeping the outer variables, f, i and the parent's positive carrier t. The whole fixed tuple gives $R \equiv 3 \pmod 4$. Write $m = \Delta I$, so $Q = m^2$, and choose

$$V = \frac{\chi_m(R)}{m}, \qquad y_{\text{old}} = \psi_m(R).$$

The parent's explicit odd-index reconstruction proves $V \equiv -c \pmod{f^2}$, $V \equiv -R \pmod c$ and $y_{\text{old}} \equiv -1 \pmod Q$. The last congruence also follows directly because $\psi_m(R)$ is an even polynomial in m with constant term $(-1)^{(R-1)/2} = -1$. Consequently

$$T = \frac{V + c + Rf^2}{cf} > 0, \qquad y = \frac{y_{\text{old}} + 1}{M} > 0$$

are integers, using $\gcd(c, f) = 1$, $f^2 \equiv 1 \pmod c$, and $M \mid Q$. The Pell identity and (*) give $Z = 0$. The other three factors are 1, so $P = 0$. This changes the meaning of y and may replace T; it does not assert the residue for every old auxiliary witness.

Exact degree. Let $q_0 = \beta J$, $Y_0 = sq_0^3$, $E_0 = wq_0Y_0$, $k_0 = \eta + \zeta$, $c_0 = k_0Y_0$, $\ell_0 = E_0c_0$, $U_0 = 4q_0^4$, $H_0 = 4E_0$, $I_0 = ic_0^2$ and $N_{t,2} = w(q_0 - F - \alpha) - zq_0$. The parent's exact cancellation identity gives degree 68 for N, with

$$N_{68} = 4U_0^4H_0^3\ell_0^3(2t - k_0).$$

The strong factor has degree 34 and leading form $-E_0^2I_0^2$. The two summands of Z each have degree at most 22, and their combined highest part is

$$Z_{22} = E_0f^2(c_0^2T^2f^2 - E_0^2y^2) \neq 0.$$

Thus $\deg P = 2 + 68 + 34 + 22 = 126$, with nonzero leading form

$$-N_{68}E_0^3I_0^2f^2(c_0^2T^2f^2 - E_0^2y^2)N_{t,2}.$$

This is a universal degree proof, not an inference from sampled rays.

Logical and reproducibility boundary. Composing both directions with the inherited compiler and syntactic calibration yields, in ZFC,

$$P \text{ has no strictly positive integer zero} \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

No global minimality or consistency decision is claimed. The literal DAG, parent and coefficient hashes, complete proof and independent finite checks accompany this section. The checker verifies polynomial identities, all live gates, the 91-operation count and actual-coefficient degree rays; these checks do not replace the unbounded compiler proof.

Attribution

The Pell, Jones, Sun and calibrated machine/compiler results are inherited. Sources46–52 contribute shared positive-coordinate words, source shears, Pell majorants, exact CRT removal, quotient absorbers and squared-modulus reconstruction. Source53 contributes selective extraction, calibrated carry blocks and seeded guards. This report's new contributions are the homogeneous-degree-separated architecture, its certified composition, the coprime spectator guard, the ordinary-root merger and explicitly audited combinations. Original notices and immutable references are preserved. AI-assisted internal reviews are not external human endorsement.